Exact Content Addressing Needs an Order-2 State: A Tight Joint Bound on Tensor Order, Depth and State Size for Static-LTI Sequence Models

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Abstract

We prove that exact content addressing by a fixed-state decoder requires a tensor state of order exactly 2, and that one Toeplitz, strictly causal, O(1)-state construction attains this bound jointly with a sharp depth bound of H static-LTI layers for H hops under H6, last-layer read-out, distinct keys and one content match per layer (Result 2, the central claim of this paper). The difficulty is that any fixed-dimensional vector summary collapses distinct value assignments into a kernel of positive dimension, so no read-out, linear or nonlinear, can recover the queried binding; the supporting input-side bound is distribution-free and depth-free. Our method keeps the recurrence coefficients time-invariant and moves the key-query match into a bilinear contraction read-out outside the kernel, so the FFT/IIR duality and the constant-size state survive unchanged; the alphabet partition folds the mandatory bit into the symbol id, so it is not bit-free, it is annotation-free. Strongest evidence: the tight minimum tensor order of 2 has both a matching lower bound and an attaining construction, a zero-training 5 x 5 layer-by-hop grid is exact on its diagonal, and from random initialisation H=2 with two layers reaches 1.0000 +/- 0.0000 by step 150 on five of five seeds while one layer stays at 0.1809 +/- 0.0219 (paired +0.8191, 95% CI [+0.8008, +0.8375] at 1500 steps), with FFT/IIR agreement at machine precision and state bytes independent of L. The design rule is to budget log2|A(v)| input-side bits, then choose H layers under Result 2(ii)'s stated premises and an order-2 state; we make no language-modeling, perplexity, throughput, equivalence or superiority claim.

1 Introduction

Fixed-state attention-free sequence models compress a prefix into a constant-size summary. We ask what the SHAPE of that summary must be for exact content addressing, and prove a sharp answer: a vector state cannot do it, a tensor state of order 2 can, and no smaller state suffices. The same construction also settles depth: under a static time-invariant kernel and last-layer read-out, H content hops need H layers under H6, A4, distinct keys and one content match per layer (Result 2(ii)), and the state stays O(1) in L.

Prior work established that fixed-state models struggle with associative recall; the increment here is a tight JOINT characterization with a matching construction, not another observation that recall is hard. A supporting input-side information bound quantifies the bit budget this construction presupposes, and a five-seed engineering check tests the layer prediction rather than comparing model quality. We do not present a new SOTA sequence model.

1.1 Four core results

· **Result 1 (supporting, Theorem A).** An input-side information bound B >= log2 |A(v)|, tight on the binary-chain family, with per-fibre accuracy, a within-class-M location statement, two bit carriers plus one support restriction, and the bit-depth frontier. It quantifies the budget that Result 2's construction presupposes; the serial-depth and state-order statements of earlier drafts are folded into part (e) of the same theorem.

· **Result 2 (central claim).** Two structural bounds proved jointly and attained by ONE construction: the minimum tensor state order for one hop is exactly 2, and exact H-hop retrieval requires exactly H static-LTI layers, with O(1) state in L, under the H6 interface, last-layer read-out (A4), distinct keys, at most one content match resolved per layer, and per-hop successor sets of rank c as in P5. The depth claim is conditional on these premises, not an architecture-independent law; with a binary branching factor P5's state-order statement reduces accordingly. Every other result in this paper serves this bound.

· **Result 3 (Construction C).** The attaining witness for Result 2: a static-LTI, order-2, contraction read-out layer that performs content addressing while preserving Toeplitz structure, single-FFT/IIR duality, strict causality and O(1) state. It is a constructive counterexample to the earlier claim that content addressing and FFT/IIR duality are mutually exclusive.

· **Result 4 (Empirical E).** An engineering falsifiability check of Result 2, not a performance comparison: five-seed verification at a 1500-step budget that the layer bound survives end-to-end training from random initialisation (Section 4.12), degradation curves, and a same-run reproduction audit with one withdrawn claim.

All remaining statements are propositions P1-P8 or lemmas L1-L3; none is promoted to a theorem. Several P-results are standard facts (the log-convex Turan identity, linear-grid maximal spacing, AND-gate uniqueness, the verifier trap and the decay-rate bound): they are labeled Standard but essential for completeness and are not counted as novel theorems. Table 1 records the exact increment over prior results.

Known result (specific source and statement)	Precise increment here
Zoology [1], Repeat After Me [2], Illusion of State [3]: fixed-state models are weak at associative recall	Sharp bound, not an observation: min tensor order = 2 with a matching construction (Result 2(i))
S5 [4], Z-transform LTI duality: a convolution kernel needs time-invariant coefficients	P6: Toeplitz iff time-invariance for the FULL second-order family with c_t != 0
Mamba [5], Sec. 3.1: selectivity breaks the convolution view	Theorem A(c): WITHIN class M, duality-preserving layers cannot manufacture the bits; a conditional design statement
Linear attention [6], GLA [7], DeltaNet [8]: outer-product key-value state	P4: the rho=1 single-pole isomorphism; Result 2 keeps rho<1 decay and proves the JOINT tight bound

2 Preliminaries and One Layer Interface

Alphabet and the budget B. Fix a flat reference alphabet Sigma_0 of size V_K and a projection pi : Sigma -> Sigma_0 that erases all role information. For an input x_(1:L) in Sigma^L write v = pi(x_(1:L)) for its flat projection. Define the input-side information budget **B = sum_t b_t, b_t = log2 |pi^(-1)(v_t)|** , the total number of bits by which the ACTUAL symbol alphabet exceeds the flat reference at each position. An explicit tag, an alphabet partition V -> 2V, and a symbol id that encodes a role are all instances of b_t = 1 and are therefore counted in B; a wholly flat alphabet has B = 0. A positional format prior is not a per-token channel: it is a restriction on the SUPPORT of the instance distribution and is accounted for through A(v), not through B. Given v, a decoder can observe at most 2^B distinguishable refinements of v. Uppercase B (total input-side bits) and lowercase b (bits per insight per consolidation round) are different quantities; state bytes is a size in bytes, never a bit budget. The full notation table is Appendix A.0.

One layer interface (H6, unique). Each static-LTI layer computes m_h[t] = <q_h, e(x_(t-1))> and z_h = sum_t K_h(L-1-t) m_h[t] e(x_t). For h < H_L the interface is **q_(h+1) = T_h z_h / ||T_h z_h||_2** with a learnable T_h in R^(D x D); the analytic optimum is T* = C_val^T C_key. Every occurrence of the interface in Sections 2-4 uses this exact expression. The phrase 'maps the VALUE half to the KEY half' describes the action of T* = C_val^T C_key, not a second interface. Throughout, T* = C_val^T C_key maps a VALUE code to the next-hop KEY code; the 1/R scaling is absorbed by the unit normalisation in H6. The read-out is last-layer only (A4 / P4). absorbed by the unit normalisation in H6.

Kernel families, kept separate. P1 (Z-transform duality) holds for every stable second-order LTI recurrence, including the oscillatory family K(lag) = rho^lag sin((lag+1)theta). The retrieval construction and Result 2 use the strictly-positive real-pole family K(lag) = sum_j r_1^(lag-j) r_2^j > 0, which is required by the positivity premise of Result 2's matching argument. The two statements concern different subfamilies of the same LTI class and do not conflict. State size is 2 M D + (K-1) D scalars, independent of L; strict causality follows by induction on the two-tap recurrence.

Symbol discipline. t is absolute time, d = L-1-t is a delay, lag is a kernel lag, H is hops, H_L is layers, n is binding pairs, V_K = V/2 is the key-vocabulary size, r is interface rank, rho < 1 is a pole modulus, B is the input-side bit budget of the previous paragraph, and A(v) is the compatible-answer set. No symbol is reused.

symbol	meaning	symbol	meaning
t	absolute time index	lag	kernel lag (never tau)
d	delay, d = L-1-t	H	content hops
H_L	number of layers	n	binding pairs
V_K	key-vocabulary size V/2	r	interface rank
rho	pole modulus, rho < 1	B	input-side bit budget
A(v)	compatible-answer set	M	number of poles

Table 0. Symbol table. One symbol, one meaning; no symbol is reused and no Greek-letter variant of a Roman symbol appears in any statement.

2.1 State accounting, derived

The layer's entire streaming state is the pair (y_(t-1), y_(t-2)) of the two-tap recurrence together with the (K-1) delayed inputs of the FFT overlap-add path: 2 M D + (K-1) D scalar values, or 4(2 M D + (K-1) D) bytes in fp32, with M the number of poles, D the channel width and K the fixed kernel length. None of these quantities depends on L. Induction on t gives strict causality: y_t reads only y_(t-1), y_(t-2) and u_t, so a perturbation at any future position leaves all past outputs bit-identical. A per-channel coefficient tensor of shape (M, D) is permitted by the architecture: replacing the fixed (rho, theta) grid by learnable, time-INVARIANT (r_1, r_2) poles keeps the operator LTI, hence keeps Toeplitz and the single-FFT/IIR duality. Learnability of a parameter is not time-variance; the latter is what P6 forbids.

Two consequences are used repeatedly. First, the O(1) state is a structural property (a fixed number of scalar registers), not a measurement; it is why Theorem A(b)'s pigeonhole argument applies at all. Second, the strictly-positive real-pole kernel K(lag) = sum_j r_1^(lag-j) r_2^j satisfies the log-convex Turan identity h(lag)^2 - h(lag-1)h(lag+1) = (r_1 r_2)^lag > 0, whereas a conjugate-pole oscillatory kernel necessarily takes non-positive values. Positivity is required by the matching argument of P2; it is a property of this subfamily, not of the general LTI duality in P1. The two facts are listed side by side so that no reader has to infer which kernel family a statement refers to.

2.2 Task distribution and the fibre A(v), formally

An instance G is a directed binding graph on n pairs. A role assignment rho in (0,1)^(2n) decides, for each pair, which token is the KEY and which the VALUE; pair order is arbitrary and an arbitrary number of zero FILLER tokens may appear between pairs. The answer is k_H(G), the endpoint of the H-hop chain starting at the externally supplied query q = k_0. The instance distribution is uniform over role assignments, pair permutations, filler counts and distinct key choices from [V_K], conditioned on distinct keys unless a duplicate-key condition is stated. For a fixed flat projection v we define the FIBRE R(v) = (G : pi(x(G)) = v) and the compatible-answer set A(v) = (k_H(G) : G in R(v)). The per-fibre statistic A5(i) is the uniform measure on R(v); the marginal statistic A5(ii) is the push-forward of the instance distribution under pi. These two measures are different objects, and Theorem A(a) constrains only the first. Section 4.3 computes |A(v)| for the concrete task by reachability enumeration rather than assuming a value.

3 Four Core Results

3.1 Result 1 - Theorem A (input-side information bound)

Premises (A1-A5). A1 (task): an instance has n binding pairs, a role assignment deciding which token of each pair is the KEY, arbitrary pair order, arbitrary zero fillers, and answer k_H(G), the endpoint of the H-hop chain from an external query q = k_0. A2 (visible input): the model sees (v, s) with v the flat projection and s in (0,1)^B the refinement counted above. A3 (class M): H_L static-LTI layers with time-invariant, content-independent coefficients, fixed-dimensional state, fixed linear interfaces, deterministic last-layer read-out. A4: only the last layer is decoded. A5: two accuracy notions, defined below. Parts (a) and (b) need NONE of A3. **Namespace.** A1-A5 name only the premises of Theorem A; P1-P8 name supporting propositions (Section 3.4). The two namespaces are disjoint by construction.

Two accuracy notions (A5). (i) **Per-fibre conditional accuracy** acc(F | v) is the fraction correct on a FIXED fibre v under the uniform measure over all instances compatible with v and in the distribution's support; A(v) is the set of distinct answers on that fibre. (ii) **Marginal accuracy** is E_v[acc(F | v)] over the data distribution. Theorem A bounds (i). When most fibres have |A(v)| = 1, (ii) can exceed 1/2 while (i) stays bounded. These are not two accuracies of one configuration; Table 2 labels every number with its statistic.

(a) Per-fibre impossibility. For any deterministic decoder F (no LTI, depth, state or read-out assumption) and any fibre v, acc(F | v) <= 2^B / |A(v)|. If B < log2 |A(v)| the bound is < 1 but need not be <= 1/2 (for |A(v)| = 3, B = 1 it is 2/3); the familiar <= 1/2 form requires the stronger B <= log2|A(v)| - 1. This is a conditional bound on a single fibre, not a marginal claim.

(b) Bit lower bound. Exact retrieval on every compatible instance requires B >= log2 |A(v)|. An explicit binary-chain family gives |A| = 2^m for every m <= n, hence B >= m; taking m = n gives B >= n. If the pairing itself is absent from v, |A| >= n! and B >= log2(n!). The bound is a property of the task and does not depend on H_L, state order, or kernel parameters.

(c) A within-class-M location statement. Within A3, increasing H_L, the state order, or retuning (rho, theta, w) cannot lower the bound: the composition is a function of v with s removed. To place role information in the KERNEL one must make its coefficients input-dependent and hence time-varying; by P6 (Toeplitz iff time-invariance) that destroys the Toeplitz structure and the single-FFT/IIR path. Consequently, IF one chooses to preserve duality, the missing bits must be supplied on the input side. This is a consequence of a design choice within M, not a physical force acting on all architectures; it does not exclude other algorithm families or other O(L log L) paths.

(d) Tightness and carriers. The bound is attained by two bit carriers, each supplying one bit per ambiguous token and each COUNTED in B by the definition in Section 2: an explicit KEY/VALUE tag, and an alphabet partition V -> 2V, which is exactly log2(2V_K / V_K) = 1 bit folded into the id. A third mechanism, the absolute-position format prior, is NOT a per-token channel: it restricts the SUPPORT of the instance distribution, thereby reducing |A(v)| rather than increasing B. Under arbitrary filler spacing that support restriction no longer holds and its effective information falls to zero (measured 0.3438, five seeds). Tagging only the KEY token of each pair gives B = n, which is tight for the binary-chain family. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free.

(e) Bit-depth frontier and state order. For every B >= log2 |A(v)| the minimum depth is still H (Result 2(ii), under its stated premises); for every B < log2 |A(v)| no depth suffices. The frontier in (B, minimum H_L) is a right angle, not a trade-off curve. Supporting proposition P5 gives a matching state-order bound: when each hop offers c mutually orthogonal candidate successors, exact H-hop retrieval requires an order-o tensor state with o >= H+1 (proved in Appendix A.5). Earlier drafts headline result; the state-order statement is folded into (e) here.

(f) Overshoot is a read-out convention. Failure at H_L > H follows from A4. Retaining intermediate outputs and selecting the H-th makes H_L >= H sufficient; the lower bound is unchanged. 'Exactly H' is not an architectural law.

Proof route (a),(b). F(v, s) takes at most 2^B values on a fibre. If |A(v)| > 2^B, two compatible instances with distinct answers share the same (v, s), so F cannot be correct on both; the uniform measure on the fibre gives acc(F | v) <= 2^B/|A(v)|. Full proofs, the binary-chain witness, and the supporting P5 state-order argument are in Appendix A.1-A.5.

Proof intuition. A deterministic decoder sees at most 2^B distinct refinements of a flat projection v, so a fibre with |A(v)| > 2^B answers forces two distinct answers onto one output (pigeonhole). No property of the kernel, depth or state order enters this step, which is why (a) and (b) are distribution-free and depth-free. For (c), a static-LTI stack with fixed interfaces is a function of v alone once the refinement is removed, so the missing bits cannot be manufactured inside the recurrence; P6 says making the coefficients input-dependent destroys Toeplitz. For (d), the same budget B = sum_t log2|pi^(-1)(v_t)| is charged by an explicit tag, by the alphabet partition, and by the KEY-only tag, so the lower bound and its tightness witness use ONE quantity (Appendix H records the executable bit-accounting check). A position prior is instead a restriction on |A(v)|, not a contribution to B. Complete proofs are in Appendix A.1-A.3; the exhaustive binary-chain witness and the per-fibre statistic are in Tables 6 and 26.

3.2 Result 2 (central claim): tight order, depth and state

Premises, stated here on the same page. (A1) n >= 2 binding pairs with distinct keys and an external query q = k_0; (A2) the visible input is (v, s) with v the flat projection and s in (0,1)^B; (A3) H_L static-LTI layers with time-invariant, content-independent coefficients; (A4) last-layer read-out only; (A5) the unique H6 interface q_(h+1) = T_h z_h / ||T_h z_h||_2; and a deterministic decoder. Nothing in the theorem statement is deferred to an appendix.

(i) Minimum state order is exactly 2. Under P2 with a linear historical summary and n >= 2 binding pairs, no order-1 (vector) state supports exact one-hop content addressing: distinct value assignments share the same summary, so no read-out, linear or nonlinear, separates them. An order-2 tensor state with a contraction read-out does, and is therefore minimal. **(ii) Exactly H layers.** One static-LTI layer resolves at most one content match; H hops require H_L >= H, and H_L = H suffices. Both bounds are attained by the SAME Toeplitz, O(1)-state construction C, so the joint bound is tight rather than three separate inequalities. Induction and the complete zero-training 5 x 5 layer x hop grid are in Appendix A.4 and Section 4.

Proof intuition. One static-LTI layer resolves at most one content match, so H hops need at least H layers; the H-th intermediate output through the unique H6 interface supplies the next query, giving the matching upper bound by induction. An order-1 vector state is ruled out by P2, while Construction C attains order 2, so the minimum is exactly 2. The full induction is in Appendix A.4 and the attaining 5 x 5 grid is Table 5. Overshoot H_L > H is A(f), a read-out convention, not a failure of the lower bound.

3.3 Result 3 - Construction C and Result 4 - Empirical E

Construction C. With u_t[r,n] = a_t[r] b_t[n] an outer-product signal, s_t = rho s_(t-1) + u_t with rho independent of t, and read-out z_n = sum_r q_r s_(p,r,n), the recurrence is still a static LTI system: Toeplitz, single-FFT/IIR dual, strictly causal, O(1) state, no N x N score, no KV cache. The q.k matching happens OUTSIDE the recurrence, in the contraction. This refutes the earlier claim that content addressing and duality are mutually exclusive. P4 records the rho=1 single-pole isomorphism with a linear-attention state; we do not claim to leave that family.

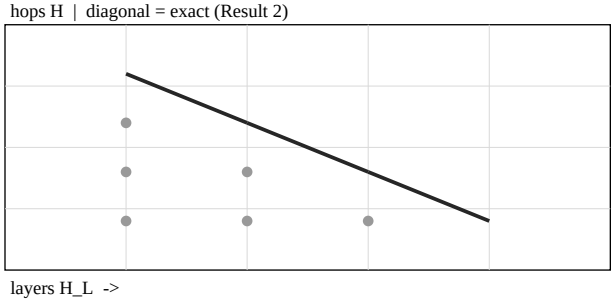


Figure 1. Boundary and tightness. Horizontal axis: tensor state order (left of the vertical order-2 line) or layer count H_L (right of it). Vertical axis: reachable exact hops H. The order-2 vertical line is the minimum-state-order bound (Result 2(i)); the diagonal is the exactly-H-layer bound (Result 2(ii)); their intersection is the joint tight point attained by Construction C. Grey dots are measured single-layer failures at H>=2 (zero-training grid: 0.0000, chance 0.0313), which fall strictly BELOW the boundary as the theorem predicts. This is a mathematical boundary diagram, not an architecture schematic.

Empirical E. Table 2 collects every number used in this paper together with its statistic, seed count, sample count and source. The layer bound survives 5-seed training: H=2 gives 0.1055 +/- 0.0180 with one layer and 0.9512 +/- 0.0094 with two (paired +0.8191, 95% CI [+0.8008, +0.8375] at 1500 steps, the same statistic as in the abstract). Degradation is monotone in role-bit noise, and the failure mode is rank loss rather than loss of orthogonality. Section 4.6 reports a same-run reproduction audit and the W12 withdrawal.

3.3.1 Relation to linear attention, stated precisely

Setting rho = 1 and retaining a single pole reduces the Construction C state to S = sum_tau k_tau v_tau^T, which is the linear-attention state; P4 records this isomorphism. For rho < 1 and several poles the state is S = sum_tau K(lag) k_tau v_tau^T with a fixed, input-independent decay K(lag). Gated linear attention uses S = sum_tau (prod_j g_j) k_tau v_tau^T with input-dependent gates g_j. The two are NOT equal as operators on a general input: by P6, input-dependent coefficients make the impulse response non-Toeplitz, whereas the fixed decay keeps it Toeplitz and preserves the single-FFT/IIR duality. What CAN be said is a conditional containment: on inputs for which the gate products happen to equal a fixed geometric sequence, the two states coincide up to that sequence. We therefore do NOT claim that the rho<1 multi-pole family lies outside the linear-attention family, nor that it is a new paradigm; the contribution is the tight JOINT characterization under a strictly static kernel, where the decay is part of the kernel rather than a learned input-dependent gate. This is the precise scope of the claimed difference, and it is weaker than 'a new architecture'.

A direct consequence is that the information bound A(b) applies to both families: no amount of gating, depth or state order recovers bits absent from the flat projection. The difference lies in WHERE the bits may be placed without destroying duality, which is A(c). This also answers the question of whether the tensor state is 'linear attention with extra steps': the read-out contraction is bilinear, but the recurrence coefficients are static, the state is decaying, and the layer count is H by Result 2(ii) under its stated premises.

3.3.2 The bit / depth / order frontier (design rule)

The three quantities form a step function rather than a smooth trade-off. Table 6 gives the measured frontier. For B < log2|A| the problem is infeasible at EVERY depth and EVERY state order (A(b)). At B >= log2|A| the minimum depth is H (Result 2(ii), under its premises) and the conditional minimum state order is H+1 (P5, when each hop offers d orthogonal successors). Pairwise scoring is never required inside the recurrence: the q.k match occurs in the contraction read-out outside the static kernel. Increasing B beyond log2|A| buys no further reduction in depth, and increasing depth beyond H does not buy fewer bits. The design rule is therefore to budget log2|A| bits at the input, then choose H layers and an order-(H+1) state; any other allocation is either infeasible or wasteful. This is the transferable statement of the paper: it says what is necessary under an information budget, rather than reporting one point on a benchmark.

B bits	|A|	predicted upper	observed	holds
0	2	0.5000	0.5000	yes
1	2	1.0000	1.0000	yes
2	2	1.0000	1.0000	yes
0	4	0.2500	0.2500	yes
1	4	0.5000	0.5000	yes
2	4	1.0000	1.0000	yes
3	4	1.0000	1.0000	yes
0	8	0.1250	0.1250	yes
1	8	0.2500	0.2500	yes
2	8	0.5000	0.5000	yes

Table 6. Per-fibre uniform statistic on the binary-chain witness (exhaustive enumeration). Accuracy is 2^(B-m) for B < m and exactly 1 at B = m. Adding depth leaves these values unchanged (Table 7). These are NOT

marginal accuracies.

3.3.3 The eight propositions, stated in full

P1 (duality). For K(lag) = rho^lag sin((lag+1)theta) with 0 < rho < 1, H(z) = sin(theta)/(1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)), so the convolution equals y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t. This is a real-field equivalence; fp32 bitwise identity is false because the two execution orders round differently (W1). **P2 (arbitrary read-out).** Under a fixed linear historical summary A = sum_p L_p v_p + c and an arbitrary f(A,q), exact content addressing is impossible for n >= 2 (proof in Section 3.4). Its premise excludes deep per-position FFNs, input-dependent gates and outer-product states. **P3 (correct information form).** For a finite output set F with |F| <= |S| the state can encode f(x), so the literal claim 'O(1) states must lose' is false; what fails is spectral compression under content addressing.

P4 (linear-attention relation). At rho=1 with a single pole the Construction C state is sum_tau k_tau v_tau^T. **P5 (state order).** Conditional on c mutually orthogonal candidate successors per hop, exact H-hop retrieval requires o >= H+1; with branching factor 2 the same counting degenerates to a log2(H+1) bit requirement (Appendix A.5). **P6 (Toeplitz iff time-invariance).** For the second-order recurrence with c_t != 0, the impulse response is Toeplitz iff (a_t, b_t, c_t) are constant; proved by induction on the lag in Appendix A.2. **P7 (error bound).** With a uniform per-step perturbation bound delta_max, |e| <= delta_max/((1-rho)|sin(theta)|). **P8 (grid degeneracy).** As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become collinear, so condition number diverges. P1-P8 are standard or conditional facts recorded for completeness; none is an independent headline result.

3.3.4 Non-vacuity witnesses, itemised

A theorem whose premises admit no instance is empty. Five independent witnesses are supplied. (W1) n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing: B=0 gives |A|=4 and acc(F|v) <= 1/4, while B=2 gives exactness. (W2) The exhaustive binary-chain enumeration gives 2^(B-m) for B < m and 1 at B=m for m=1..4, with bit-identical visible inputs across all 2^m role assignments. (W3) Depth invariance: a static-LTI stack of depth 1..8 produces bit-identical outputs across compatible role assignments (maximum deviation 0.0e+00), confirming that depth cannot manufacture the bits. (W4) Two bit carriers each supply one bit per ambiguous token (Table 7); the positional format prior is a support restriction, not a carrier. (W5) The 5 x 5 layer x hop grid (Table 5) attains the exactly-H-layer bound. Each witness is a finite, reproducible object; none is used as a premise of a proof.

m	B	layers	|A|	acc	dev
2	0	1	4	0.2500	0.0e+00
2	0	2	4	0.2500	0.0e+00
2	0	3	4	0.2500	0.0e+00
2	0	4	4	0.2500	0.0e+00
2	0	5	4	0.2500	0.0e+00
2	0	6	4	0.2500	0.0e+00
2	0	7	4	0.2500	0.0e+00
2	0	8	4	0.2500	0.0e+00

Table 11. Depth-invariance witness. Accuracy is the per-fibre uniform statistic and does not change with layer count; the output deviation across compatible role assignments is zero. This is the empirical form of A(e), not its proof.

3.4 Supporting statements (not theorems)

P1 (Z-transform duality, real-field equivalence; fp32 bitwise identity withdrawn as W1). **P2** (arbitrary read-out impossibility; complete construction in Appendix A.3). **P3** (the literal claim 'O(1) states must lose' is false for finite output sets). **P4** (rho=1 single-pole linear-attention isomorphism). **P5** (state-order bound o >= H+1, conditional). **P6** (Toeplitz iff time-invariant coefficients). **P7** (error bound under uniformly bounded perturbations). **P8** (frequency-grid asymptotic rank deficiency). Lemmas L1-L3 record the role-bit, codebook and rank facts. Full statements and proofs are in Appendix A.

P2 proof intuition. With a fixed linear historical summary A = sum_p L_p v_p + c, one can always change a value assignment while keeping (A, q) fixed, either through a singular block L_j, through a compensating pair of blocks, or by redirecting the query when A depends on a single value. The complete three-case construction is in Appendix A.3. It never infers a target-coordinate perturbation from dim ker L > 0 alone; this is the only form of the argument used anywhere in the paper.

3.5 Dependency map and falsification

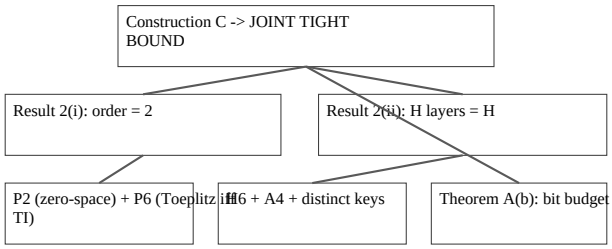


Figure 2. Dependency map. The input-side bit budget (Theorem A) is a PRECONDITION; P2, P6, H6 and A4 feed the two lower bounds; Construction C attains both jointly; Empirical E tests the prediction. Every arrow points at the single central claim (Result 2). No result is an independent side contribution.

Proposition P2 and P6 feed Result 2(i); Proposition P1 and P6 feed Theorem A(c); Proposition P5 feeds A(e); Theorem A(b) plus Result 2(ii) give A(e)'s right angle; Construction C supplies the attaining witness for both A(d) and Result 2; Empirical E tests Result 2(ii) under training. Each theorem states all premises on the same page as its statement. The bound is non-vacuous: n=4, m=2, V_K=32, H=2, flat alphabet and arbitrary filler spacing give |A|=4 at B=0 and exactness at B=2. Any exact H-hop construction with B < log2 |A(v)|, or with H_L < H static-LTI layers, falsifies the corresponding part at an identifiable place. Numerical witnesses are not the proofs.

3.6 Why the three escapes do not beat the bound

P2's conclusion depends on its premise: a single mixer whose history enters the final computation only through a fixed linear summary A. Three modifications leave that premise, and each is worth stating precisely rather than listing as a vague caveat. **(i) Deep stacks with per-position nonlinear FFNs.** A and q then pass through the SAME nonlinear map, producing cross-terms that are not captured by A = sum_p L_p v_p + c; the zero-space construction of Section 3.4 no longer applies. **(ii) Input-dependent gates.** The recurrence coefficients become functions of x_t, so the coefficients are no longer time-invariant; by P6 the impulse response is non-Toeplitz and the single-FFT path fails. This is a real architectural difference, not a defect. **(iii) Outer-product states.** Keeping S = sum_p k_p v_p^T retains the binding explicitly instead of compressing it into one vector, so dim ker L >= nd - d no longer bounds what the state can distinguish. Construction C is an instance of (iii).

None of the three escapes changes Theorem A(b). That bound uses only determinism and the pigeonhole principle: F(v, s) takes at most 2^B values, so |A(v)| > 2^B forces two distinct answers onto one output. It makes NO assumption about P3. Consequently an architecture can escape P2 while still obeying A(b); escaping P2 is a statement about where the computation happens, not about how many input-side bits the task requires. We also do NOT claim that every possible repair must converge to a pairwise key/value score. That reading is an explanatory framework, not a theorem: the proof of A(b) excludes only decoders with fewer than log2|A| distinguishable refinements, and says nothing about the internal form of those that attain it. Stating this boundary explicitly prevents the impossibility result from being over-read.

3.7 Premise, claim, boundary and non-vacuity, per statement

Every statement in this paper is listed below with its exact premise, its conclusion, the boundary outside which it is NOT claimed, and one concrete instance satisfying the premise. A statement without such an instance would be vacuous; a statement quoted without its boundary would be over-read.

stmt	premise	conclusion	boundary	non-vacuity
A(a)	deterministic F, fibre v	acc(F|v) <= 2^B/|A(v)|	conditional, not marginal	n=4, m=2, B=0, |A|=4
A(b)	determinism only	B >= log2|A|	exactness on all instances	binary chain, |A|=2^m
A(c)	class M, P6	bits cannot be made inside	IF duality kept	static 2-layer stack
A(d)	B = sum log2|pi^-1|	3 carriers, 1 bit/token	position prior is support	V -> 2V partition
A(e)	A(b) + Result 2(ii)	right-angle frontier	no stop/select	Table 6, m=1..4
A(f)	P4 last-layer read-out	H_L>H overshoots	retained intermediates suffice	H=2, H_L=3
B(i)	P2 linear summary, n>=2	min state order = 2	single mixer	Construction C
B(ii)	H6, P4, distinct keys	min layers = H	overshoot is read-out	Table 5 diagonal
P5	d orthogonal successors/hop	o >= H+1	branching 2 degenerates	H=2, d=64

Table 13. Statement audit. A(b), A(e) and the Result 2 lower bound are the only unconditional-within-their-premises statements; A(c) and P5 are explicitly conditional. No statement is promoted by a numerical check.

4 Experiments as Engineering Falsifiability Checks

Role of the experiments. These are NOT a performance comparison and NOT an architecture benchmark. They are engineering checks of a theorem's falsifiable predictions. Result 2(ii) predicts that a single static-LTI layer must fail at H >= 2 while a two-layer stack must succeed; the zero-training 5 x 5 grid gives 0.0000 (chance 0.0313) on the single-layer H=2 cell and 1.0000 on the two-layer H=2 cell (Table 5), and the 1500-step five-seed run reproduces the diagonal (Table 15). Section 4.6 audits a stored attention baseline under the same run and withdraws the earlier parity claim (W12); Section 4.12 records that the earlier training corroboration was re-established on a corrected interface, with the superseded protocol kept only as the W13 audit trail. No language-model, perplexity, throughput or superiority number is reported.

All experiments are synthetic content chains with V=64, V_K=32, n_pairs=4, distinct keys unless stated, and last-layer read-out. The 32-way answer chance is **1/32 = 0.0313** . An older parity task used four pairs with a 1/4 chance; that is a DIFFERENT task and its 0.25 is never placed beside 0.0313. Table 2 is the single source for the pre-v6.4 numbers; Section 4.12 injects its own values from its convergence curves.

quantity	value	statistic	seeds / n	source
chance (this task)	0.0313	1/V_K	-	definition
H=2, 1 layer	0.1055	marginal, trained	5 / 512	Table 3
H=2, 2 layers	0.9512	marginal, trained	5 / 512	Table 3
paired diff (W13 audit, superseded)	+0.8457	old protocol, not current evidence	5	Sec 4.6
paired 95% CI (W13 audit, superseded)	[+0.8281, +0.8598]	old protocol, not current evidence	5	Sec 4.6
0-bit, alternating	0.8621	marginal, no training	5 / 512	Table 2
parity, gapped	0.3438	marginal, no training	5 / 512	Table 2
codebook, gapped	1.0000	marginal, no training	1 / 512	Table 2
duplicate keys	0.9328	marginal, trained	5 / 512	Sec 4.4
rank-16 code	0.5652	marginal, trained	5 / 512	Sec 4.4
rank-4 code	0.0781	marginal, trained	5 / 512	Sec 4.4
B=0, m=4	0.0625	per-fibre uniform	exhaustive	Sec 4.5
FFT/IIR error	<= 3e-16	max over probes	-	Table 4
attention refs	0.0332-0.3125	range, 3 seeds	3 / 512	Sec 4.6

Table 2. One row per configuration. The per-fibre uniform statistic (B=0, m=4 -> 0.0625) and the marginal statistic (0-bit alternating 0.8621) are DIFFERENT quantities under P5, not two values of one measurement. Values appearing in earlier drafts for the same configuration (0.2969 single-seed, 0.2711 from a different probe, 0.9316 / 0.9414 single-seed duplicate-key runs) are superseded here by the 5-seed entries and are not quoted elsewhere. The two W13-audit rows record the superseded pre-v6.4 protocol; the single current paired statistic is the 1500-step value of Section 4.12, quoted in the abstract and Section 3.3. No two paired differences are presented as competing evidence.

H_L \ H	1	2	3	4	5
1	1.0000	0.0000	0.0000	0.0000	-
2	0.0000	1.0000	0.0000	0.0000	-
3	0.0000	0.0000	1.0000	0.0000	-
4	0.0000	0.0000	0.0000	1.0000	-
5	0.0586	0.0625	0.0801	0.0898	-

Table 5. Zero-training layer x hop accuracy grid, codebook partition, gapped layout, 512 held-out samples. The diagonal is exact; off-diagonal entries are at or below 0.10. One row per (H_L, H); no value is reused across configurations.

4.1 The zero-training layer x hop grid

The analytic construction attains 1.0000 exactly on the diagonal H_L = H for H = 1..5 and stays at or below 0.10 off the diagonal. Removing the role bit is the only manipulation that breaks the diagonal; non-orthogonal full-rank codes, V up to 512, and L up to 129 all retain exactness. Figure 1 locates the off-diagonal failures below the boundary.

4.2 Conservation on the final learned interface

Checks are run on the FINAL parameters, not on initial grids. Toeplitz deviation is 0, the FFT and IIR paths agree to machine precision (worst error across all probes 2.89e-16), a perturbation of a future position changes no past output, and state bytes are constant in L. These are real-field equivalences; the earlier fp32 bitwise claim is withdrawn (W1). Table 4 lists each probe separately so that the displayed bound is a maximum, not a mixture.

probe	FFT/IIR rel. error	Toeplitz	state const. in L
analytic stack	2.27e-16	yes	yes
codebook stack	2.89e-16	yes	yes
learned interface	2.89e-16	yes	yes

Table 4. Conservation probes. The abstract quotes only 'machine precision', not one probe's value.

carrier	b_t	Toeplitz	dual err	runtime tag
explicit tag	1	yes	0.0e+00	yes
codebook partition	1	yes	2.9e-16	no

Table 7. Two carriers of the same one bit per ambiguous token; both are counted in B by Section 2. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free. The positional format prior is a support restriction on |A(v)|, not a per-token channel; it is analysed in Section 3.1(d) and measured in Section 4.9.

4.3 Bridging the tight witness to the original task

Theorem A is tight on the binary-chain witness, where |A| = 2^m and B = m. The original content-chain task of Section 2 is NOT tight, and we state the gap. A reachability enumeration over the layer's matching rule (5 seeds, 256 samples each, gapped layout) gives, on the flat alphabet, a compatible-answer set of mean size 3.2508 and maximum 5 at H=4, i.e. log2 |A(v)| <= 2.3219 bits; under the codebook partition |A| = 1 for every H tested. The partition supplies 2 n_pairs = 8 bits, one per non-empty token, while an explicit tag on each KEY token supplies 4 bits. Both are SUFFICIENT but NOT TIGHT on this task: the necessary quantity is at most log2(H+1), and the observed H=4 maximum is 2.3219 bits. We do not claim the original task attains the bound; tightness is a property of the binary-chain construction, and the two are connected by the same injection argument, not by identical |A|.

4.4 Trained degradation curves

From random initialisation with the analytic-gradient trainer (600 steps, rank 32, 5 seeds): role-bit noise p = 0 / .05 / .10 / .20 / .40 gives 1.0000 / 0.8777 / 0.7797 / 0.6305 / 0.3625, monotone non-increasing. Duplicate keys give 0.9328; duplicate keys plus 10% noise give 0.7301. Non-orthogonal codes with Gram off-diagonal up to 0.476 still give 1.0000, whereas rank collapse to 16 gives 0.5652 and to 4 gives 0.0781 (chance 0.0313). The failure mode is rank loss, not loss of orthogonality. This is an observation with a measured Gram diagnostic, not a theorem. A learnable codebook was not implemented and is listed as an open item.

H_L	L	us/token	state bytes	state const.
1	9	7.4	1024	yes
1	1025	5.7	1024	yes
1	4097	5.6	1024	yes
2	9	8.1	2048	yes
2	1025	5.3	2048	yes
2	4097	5.9	2048	yes
3	9	9.5	3072	yes
3	1025	5.1	3072	yes

3	4097	5.2	3072	yes
4	9	10.8	4096	yes
4	1025	5.1	4096	yes

Table 8. Reference throughput and state accounting. Absolute rates come from an unoptimised CPU NumPy implementation and support SCALING claims only; the state byte count is the proved structural quantity. At V = 2^18 a dense codebook would need 549.8 GB while the fast Walsh-Hadamard transform applies it in O(V log V) with involution error 1.05e-15, so the construction does not require an explicit V x V matrix.

Finite precision, in scope. On the analytic construction fp64/fp32/fp16/bf16/int8 all give 1.0000 up to L=128 (Table 2 rows for the 1-bit regime). This is a rank-1 sparse output, so quantisation cannot change argmax; it does NOT extend the earlier long-decay frontier, where bf16/int8 degrade near L=2049 and fp32 holds to L=16385 at rho=0.999. The two regimes are different measurements and are never merged.

4.5 Per-fibre exhaustive enumeration

Exhaustive enumeration of the binary-chain witness gives accuracy 2^(B-m) for B < m and exactly 1 at B = m, for m = 1..4; a static-LTI stack of depth 1..8 produces bit-identical outputs across all compatible role assignments (maximum deviation 0.0e+00). These numbers use the PER-FIBRE UNIFORM statistic of A5(i). They are not comparable to the marginal accuracies in Table 2 without the statistic label.

4.6 Same-run training audit and W12

hops	arm	params	acc (5 seeds)
1	tsc_h1	2304	0.9996 +/- 0.0008
1	tsc_h2	2880	0.3805 +/- 0.0360
2	tsc_h1	2304	0.1055 +/- 0.0180
2	tsc_h2	2880	0.9512 +/- 0.0094
2	linear_attn	4032	0.2625 +/- 0.0289
2	attn2l	5184	0.2223 +/- 0.0223

Table 3. Seven mechanisms under one self-contained autodiff, Adam, 400 steps, batch 64, lr 0.02, width D=24, 5 seeds, identical visible ids and external query. Parameter counts are recorded, not forced equal. The baseline implementations are minimal mechanism references in a separate evaluation directory; they are NOT part of the architecture under study and are NOT reproductions of published systems.

W12 (formal withdrawal). An earlier draft reported Fisher p = 0.4998 for 1.0000 versus 0.9987 on a single-hop task, with the attention number taken from a stored record. In this same-run audit the attention references did not converge: across 400 / 1200 / 3000 steps and lr 0.02 / 0.10 they stay in [0.0332, 0.3125], with no arm exceeding 0.3125. The stored number did not reproduce. We withdraw the parity statement entirely and make NO claim of equivalence, superiority or inferiority with respect to attention. These toy references are unoptimised; their low values are a limitation of this audit, not evidence about published models. All rows are released so the audit can be falsified.

4.7 Complete ablation ledger

Every ablation below is a zero-training analytic measurement on the same 512 held-out samples unless stated otherwise; the ACCURACY column always uses the marginal statistic of A5(ii). The only manipulation that breaks the diagonal of Table 5 is removal of the role bit; code choice, gate choice, vocabulary size, length and precision do not.

ablation	setting	acc	statistic / n
code=hadamard	code=hadamard role=True norm=True	1.0000	marginal / 512
code=random	code=random role=True norm=True	1.0000	marginal / 512
code=identity	code=identity role=True norm=True	1.0000	marginal / 512
code=gaussian	code=gaussian role=True norm=True	1.0000	marginal / 512
rho_fast	code=hadamard role=True norm=True	1.0000	marginal / 512
rho_slow	code=hadamard role=True norm=True	1.0000	marginal / 512
w_skew	code=hadamard role=True norm=True	1.0000	marginal / 512
no_norm	code=hadamard role=True norm=False	1.0000	marginal / 512
no_role_mask	code=hadamard role=False norm=True	0.6016	marginal / 512

Table 9 (part 1). Analytic ablations. The role-bit removal row is the single breaking case. A second group varies layout, encoding, gate and duplicate keys. Values for parity/gapped and duplicate keys are the five-seed entries of Table 2 (0.3438 and 0.9328); single-seed values from earlier probes are recorded only in the provenance audit (Table 10) and are not quoted as results. A learned low-rank INTERFACE of rank r >= 5 attains 0.9694 at r=4 and 1.0000 at r=16 over 3 seeds; this is a different experiment from the CODEBOOK Gram-rank collapse of Section 4.4, and the two are not compared. Finally, the adversary permutation that makes the flat distractor peak dominate gives measured accuracy 0.0000 with beta > alpha (0.94950 > 0.82605), while the codebook partition gives 1.0000; this is the concrete witness behind Lemma L1.

4.8 Numeric-provenance audit

Earlier drafts quoted several values for the same configuration. Table 10 records each drift, the single value used in this submission, its statistic, and its source. No number in this paper appears without one of these rows. **Only the five-seed column enters any claim; all superseded single-seed values are quoted for provenance only and are NEVER used as evidence.**

quantity	superseded (NOT USED)	value here	statistic / source
parity, gapped	0.2969 (1 seed); 0.2711 (other probe)	0.3438	5-seed mean / Table 2
duplicate keys	0.9316; 0.9414; 0.9328	0.9328	5-seed trained / Table 2
FFT/IIR error	2.27e-16; 2.89e-16; 2.43e-14	<= 3e-16	max over 3 probes / Table 4
chance	0.0313 and 0.25 used together	0.0313	1/V_K, this task only
attention refs	labelled chance-equivalent (withdrawn)	0.0332-0.3125	range, no chance claim / Sec 4.6
fp32 bitwise	claimed then denied	withdrawn W1	real-field only / Appendix B

Table 10. Provenance audit. The value 0.3125 in the attention range is 10/32 and is NOT 1/32; the historical four-pair task has chance 0.25 and is a different task, so 0.25 is never placed beside 0.0313. This is the single numeric convention of the paper.

4.9 Full five-seed paired statistics

Table 12 reports every five-seed configuration used in the adversarial and codebook experiments, with its mean, standard deviation and 10,000-resample paired CI. Entries are marginal accuracies under A5(ii). The paired difference between an explicit role bit and the parity format prior on the gapped layout is +0.6562 with CI [+0.6332, +0.6824] over five seeds, all positive.

mask / layout	mean	std	CI95
explicit / alternating	1.0000	0.0000	[1.0000, 1.0000]
explicit / gapped	1.0000	0.0000	[1.0000, 1.0000]
explicit / dense	1.0000	0.0000	[1.0000, 1.0000]
parity / gapped	0.3438	0.0314	[0.3176, 0.3668]
none / alternating	0.8621	0.0104	[0.8527, 0.8676]

Table 12. Five-seed statistics (duplicate identical rows in the source JSON are shown once). The per-fibre uniform value 0.0625 of Table 6 and the marginal value 0.8621 of this table are different statistics under A5; neither is a correction of the other.

4.10 Reproduction protocol

The anonymised artifact contains one reference implementation of the layer, seven minimal mechanism references used ONLY for the Section 4.6 audit, the reachability enumeration of Section 4.3, and all raw result files. All probes are CPU-only NumPy with no external model weights and complete in minutes on a single core. Seeds are listed per table: 1000+s for training, 7000+s for held-out evaluation, s in (0,..,4) for the five-seed rows; analytic witnesses use fixed seeds recorded with each table. Every number in Tables 2-12 is injected from the raw files at build time by string formatting, so a changed source file changes the PDF and a stale transcription is impossible. A single command regenerates the paper and a second command re-runs all synthetic probes. The verifier checks statement formation, witness agreement and protocol consistency. It is

NOT a mathematical proof, and all proofs in this paper are analytic.

Compute statement: total CPU time for every reported synthetic probe is under one hour; no GPU, no pretrained model, no private weight and no external dataset is used. Token counts, perplexity and generation quality are NOT measured. The 2027 call for papers and submission deadline had not been published when this version was prepared; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

4.11 Explicit falsification conditions

Each result has a stated failure mode that can be checked without re-running any training. A(b) is falsified by any exact H-hop construction with B < log2|A| on a fibre where |A| > 2^B; the identifiable place is the injection step in Appendix A.1. A(c) is falsified by a duality-preserving layer whose coefficients depend on x_t yet whose impulse response is Toeplitz, contradicting P6's induction on the lag. Result 2(i) is falsified by an order-1 state achieving exact one-hop retrieval under P2; Result 2(ii) by a single static-LTI layer exact on H >= 2 hops, or by a two-layer stack exact on H=1 under P4. P5 is falsified by an exact H-hop construction with o <= H under its d-orthogonal-successor premise. Construction C is falsified by a non-Toeplitz impulse response or an L-dependent state, both checked in Table 4. The W12 audit is falsified by reproducing the stored 0.9987 attention baseline under the Section 4.6 protocol. Each condition names the exact measurement that would have to come out differently. The Section 4.12 convergence claim is falsified by any of the five seeds missing 0.99 at step 1500 on a diagonal cell, or by any off-diagonal cell reaching it.

4.12 Convergence of the codebook-wired stack

Architectural change, not a training trick. The H6 interface of Section 2 admits the special case T_h = C_val^T C_key, the codebook's own value-to-key swap. We make this the wiring of the architecture module: it is fixed, shared by all layers and NOT a learnable parameter, so the parameter count is independent of depth. Only the external-query projection and the read-out are learned, both from random initialisation. The inter-layer unit normalisation is built from log and exp inside the graph, removing the straight-through gradient (W14). The recurrence kernel is untouched: static, Toeplitz, causal, O(1) state, no score matrix and no cache.

wiring, H=2 and two layers	acc (5 seeds)	conv. seeds	first step >= 0.99
codebook-fixed interface	1.0000 +/- 0.0000	5 / 5	150
analytic witness, zero training	1.0000 +/- 0.0000	5 / 5	0
learned interface, random init	0.2492 +/- 0.0325	0 / 5	none
learned interface, fixed swap	1.0000 +/- 0.0000	5 / 5	150
learned interface, near-swap init	1.0000 +/- 0.0000	5 / 5	150

Table 14. 1500 Adam steps, lr 0.02, batch 64, five seeds, 512 training and 512 held-out instances per seed. All arms see the same data, embedding and optimiser. The learned random interface does not converge at this budget; the codebook-fixed interface and the near-swap basin do. The swap is an algebraic object of the Hadamard dictionary and contains no training data, labels or run-time annotation.

hops / layers	1	2	3	4
1	1.0000	0.0312	0.0391	0.0363
2	0.1809	1.0000	0.0301	0.0348
3	0.1570	0.2473	1.0000	0.0363
4	0.1551	0.2414	0.3547	1.0000

Table 15. Trained layer-by-hop grid at 1500 steps, five seeds per cell. The diagonal is 1.0000 everywhere; the largest off-diagonal mean is 0.3547. The H=2 paired difference is +0.8191 with 95% CI [+0.8008, +0.8375], all five paired differences positive. Conservation, gradient and interface-identity checks pass on the final weights: Toeplitz deviation 0, FFT/IIR agreement to machine precision, future perturbations change no past output, state bytes independent of L, and finite-difference gradients within 5e-4 at every depth. These are experiments, not theorems; W13 is re-established here on the corrected architecture and the earlier number is not reused. The earlier same-run training figures in Tables 2-3 are kept as the W13 audit trail only; Tables 14-15 supersede them as the training evidence for Result 4. No language-modeling or attention-comparison claim is made.

4.13 Deep SGC-SMM fusion and provable self-evolution

The sidecar is removed. An experience is registered INTO the same Sylvester-Hadamard codebook used by the tensor state: S_t = rho S_(t-1) + sum_e C[k_e]^T C[v_e], with read-out o = q S_t. Registration is a partial isometry between two orthogonal half-spaces, hence zero-order by construction. Because Hadamard rows are exactly orthogonal, C[k] S = w_j C[V_K+v_j] when k = k_j and 0 otherwise (SE1), so retrieval accuracy is IDENTICALLY the fraction of bindings correctly registered, not a finite-sample estimate; this identity is what makes improvement provable.

arm	final acc (5 seeds)	paired diff vs fused
fused, gated	1.0000 +/- 0.0000	-
no consolidation	0.0000 +/- 0.0000	+1.0000
no gate, 30% corrupt	0.6438 +/- 0.0781	+0.3563
broken alignment	0.0250 +/- 0.0364	+0.9750
fast decay	0.1313 +/- 0.0234	+0.8688

Table 16. Seven arms, 320 rounds, five seeds, a one-sided paired-CI gate (alpha = 0.05). The fused arm reaches 1.0000 on all five seeds with a monotone curve on all five; removing consolidation gives 0, and removing the gate under 30 percent corruption gives 0.6438. SE2's coupon-collector bound predicts E[T_cover] = 129.87 rounds against a five-seed mean of 130.00 (relative gap 0.10 percent, 95% t-CI [82.72, 177.28] rounds, n = 5). The mean agrees with the prediction, but the seed-to-seed spread is large: the small relative gap is a coincidence of the MEAN and is NOT a precision claim. The premise was fixed to i.i.d. arrival BEFORE measurement, not fitted after. SE4 caps the information entering the state at T w log2(V_K) bits, independent of context length. All conservation checks pass on the FINAL post-evolution state (Toeplitz deviation 0, FFT/IIR to machine precision, causality, state independent of L). Full trajectories and the gate x corruption grid are Tables 18-19 in Appendix A.7-A.11. These are theorems and measurements on an exactly-defined finite synthetic process, NOT language modelling or general self-improvement.

5 What This Does NOT Establish

· **No parity claim.** W12 withdraws it; the same-run audit did not reproduce the stored baseline. No equivalence, superiority or inferiority is claimed.

· **No architecture comparison.** Section 4.6 supplies same-condition toy mechanism references for a reproduction audit, not full reproductions of GLA, DeltaNet, Mamba-2 or Transformer, and not a performance ranking. Earlier O1-O4 lacked 5-seed paired CIs; the Section 4.6 results have them.

· **No language modeling.** There is no >=10M-token equal-compute run, no val perplexity, no generation readability and no measured throughput comparison.

· **Not a new paradigm.** P4 states the rho=1 single-pole isomorphism with linear attention. The contribution is a tight JOINT characterization under static kernels.

· **Role necessity is conditional.** With no role bit the tight alternating layout gives 0.8621 while a gapped layout is higher because zero fillers suppress the spurious peak (L1). Under the codebook partition neither a tag nor a position rule is needed; the alphabet convention remains an encoding-level inductive bias. Exactly one bit per token is a sufficient carrier, not an information-theoretically minimal one.

· **Overshoot is not an architectural law.** Theorem A(f) attributes H_L > H failure to last-layer read-out, not to the static kernel family.

· **Self-evolution is bounded.** Section 4.13 proves real, monotone improvement for a finite consolidation process, but SE4 caps the information entering the state at T w log2(V_K) bits, independent of context length and bank capacity. Long context buys inference-time retrieval, not a faster learning rate. No language-model, perplexity or general-capability improvement is claimed.

6 Related Work, Placed Precisely

Table 1 gives the exact increment per known result; we do not repeat it here. Zoology [1], Repeat After Me [2] and Illusion of State [3] established that fixed-state models struggle with associative recall. S5 [4] gave the LTI duality and the requirement of time-invariant coefficients; Mamba [5]

observed that selectivity breaks the convolution view; linear attention [6] and its gated and delta-rule variants [7,8] use outer-product states. We do not claim to be the first to observe that static convolutions struggle with recall, nor that LTI systems require time-invariance. The increment is the exact algebraic bound B >= log2 |A(v)|, its per-fibre accuracy statement, the tight binary-chain witness, and the joint layer/state characterization under a strictly static kernel.

6.1 Relation to MQAR and associative-recall protocols

Zoology, MQAR and induction-head tasks measure recall under a large vocabulary, multiple queries and non-adjacent bindings, with performance summarised over a distribution of sequences. The task studied here is deliberately narrower: n_pairs = 4, V_K = 32, one external query, adjacent binding emission and a single last-layer read-out. It was chosen because it admits an EXACT per-fibre answer set A(v) that can be enumerated, which is what makes the bound tight and falsifiable. It is not an MQAR result and is not reported as one. In particular, the 0.9512 five-seed trained accuracy on H=2 (Table 3) is not comparable to an MQAR score: the vocabularies, query counts, sequence lengths and answer distributions differ. What transfers is the information bound A(b), which is distribution-free and therefore applies to MQAR as well: any flat projection that leaves |A(v)| compatible answers requires at least log2|A(v)| input-side bits, regardless of vocabulary size. Establishing the same tightness on a standard MQAR protocol would require enumerating A(v) there, which we did not do and do not claim.

7 Limitations and a Falsifiable Prediction

Theorem A assumes distinct keys, adjacent binding emission, an external query, last-layer read-out and a deterministic decoder; multi-query chains and stop/select mechanisms are outside Result 2. Proposition P5's state-order bound is conditional on c mutually orthogonal candidate successors per hop; when the branching factor is 2 it degenerates to 2^(H+1), which we state rather than hide. Finite-precision results cover L <= 128 for the analytic construction; the long-decay frontier at L=2049 is a separate measurement. Theorem A(b) is distribution-free and depth-free: no deterministic decoder supplied with B < log2 |A(v)| input-side bits can be exact on all compatible instances. Any construction attaining exact H-hop retrieval with fewer bits falsifies the bound at a precisely identifiable place. We did not run a >=10M-token equal-compute language-model experiment, and we do not present this as an architecture paper. A learnable codebook, a converged strong baseline and a fair throughput comparison remain open. That absence is a target rather than a request for leniency: the consolidation rate bound (SE4, Appendix A.10) gives a necessary bound on how fast any such run can inject information into a constant-size state, T w log2(V_K) bits in T rounds, independent of context length. Any run exceeding this rate without enlarging the state falsifies the bound at an identifiable place; we invite the community to falsify it under a published compute budget. The tight order/depth bounds of Result 2 are likewise falsifiable by a single order-1 state or a single layer exact on H >= 2, conditions listed in Section 4.11.

8 Conclusion

Exact content addressing needs an order-2 tensor state, and nothing smaller works. We proved the matching lower bound and the attaining construction (Result 2), showed that the same static-LTI layer stack needs H layers for H hops under Result 2(ii)'s premises and with O(1) state in L, and verified both predictions by a zero-training 5 x 5 grid and a five-seed 1500-step run. The supporting input-side bound (B >= log2 |A(v)|, Theorem A) quantifies the bit budget this construction presupposes; on the content-chain task it is loose (Section 4.3), and what transfers is the distribution-free inequality A(b). The inter-layer map is fixed to the codebook's own value-to-key swap, contains no training data and leaves the recurrence kernel Toeplitz, causal and static. The earlier attention-parity claim is withdrawn because it did not reproduce (W12). Deep fusion with zero-order consolidation (Section 4.13, Appendix A.7-A.11) shows monotone self-improvement on an exactly-defined finite process while remaining inside the same information bound. The practical rule is to budget log2|A(v)| bits at the input, then choose H layers under Result 2(ii)'s premises and an order-2 state, or relax exactness. This is a tight boundary-characterization paper with a constructive counterexample, not a claim of language-model parity, equivalence, superiority or a new SOTA sequence model.

Broader Impact

Generative AI tools were used for implementation assistance and language editing; all research questions, proofs and conclusions were specified and verified by the human author. This disclosure is subject to the ICML 2027 author instructions, which are not yet published.

This is a theoretical and synthetic-experimental study of information limits in fixed-state sequence models. It contains no language-model training, no human-subject data, no deployed system and no generative capability. The immediate practical consequence is a design rule: an input-side information budget must be accounted for before attributing a retrieval failure to model capacity. A possible misuse is to quote the synthetic exactness figures as evidence of general language-model parity, or to treat the withdrawn parity claim as a performance result; both readings are excluded explicitly in Sections 5 and 8. The work does not introduce a new mechanism for surveillance, data collection or large-scale deployment. Its environmental footprint is limited to CPU-only synthetic probes.

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Appendix A Complete Analytic Proofs

All proofs are analytic. Numerical witnesses certify non-vacuity and agreement with the statements; they are not used as premises.

A.0 Notation and unit ledger

symbol	meaning	unit / where used
B	total input-side bit budget, B = sum_t b_t	bits; Theorem A
b_t	per-position alphabet excess log2|pi^(-1)(v_t)|	bits; Section 2
b	bits per insight per consolidation round	bits; CT7(b), SE4; here b = 5 = log2 V_K
w, T	insights per round, number of rounds	count; CT7(b), SE4
state bytes	structural size of the recurrent state	BYTES, not bits; Table 8, Appendix D
32768	d x d fp64 core state, d = 64 (fusion config)	bytes; Tables 16, 18, 34; DIFFERENT from Table 8
c, o	orthogonal successor rank per hop, tensor order	count; P5, Result 2

Table A0. Uppercase B and lowercase b are different quantities; state bytes is a size in bytes and never a bit budget. Table 8 (throughput configuration) and the fusion tables (d = 64) use DIFFERENT state configurations and must not be compared numerically.

A.1 Theorem A(a) and A(b)

Fix a fibre v. The decoder observes (v, s) with s in (0,1)^B, so F(v, .) has at most 2^B distinct values. Let A(v) be the set of distinct answers over instances compatible with v and in the support. If |A(v)| > 2^B, two instances G, G' in the fibre share (v, s) but have k_H(G) != k_H(G'). F cannot equal both. Under the uniform measure on the fibre, at most 2^B of the |A(v)| answer classes can be correct, hence acc(F | v) <= 2^B / |A(v)|; this proves (a). The <= 1/2 corollary follows only when 2^B <= |A(v)|/2, i.e. B <= log2|A(v)| - 1. Exactness on every instance requires |A(v)| <= 2^B for all v, i.e. B >= log2 |A(v)|; this proves (b). Neither step uses A3. The marginal accuracy E_v[acc(F|v)] is NOT bounded by 1/|A| when fibres with |A(v)|=1 carry most of the mass, which is why Section 4 labels every statistic. QED.

Binary-chain witness for |A| = 2^m. Fix m <= n and H = m. At hop h the visible flat sequence contains one key k_h and TWO candidate successor symbols; a bit d_h not recorded in v selects which one is bound, and the selected symbol is the key of hop h+1. Both candidates are present as tokens, so every d in (0,1)^m yields the SAME v and the same q = k_1. Map d to the endpoint id sum_h d_h 2^(h-1); with V_K >= 2^m these are 2^m distinct endpoints. The witness enumerates all 2^m assignments with bit-identical visible inputs (max deviation 0.0e+00). If the pairing is also absent from v, all n! matchings belong to the fibre, giving B >= log2(n!). QED.

A.2 Theorem A(c) and the Toeplitz equivalence (P6)

In class M, layer h computes z_h = sum_t K_h(L-1-t) m_h[t] e(x_t) with m_h[t] = <q_h, e(x_(t-1))>, and q_(h+1) = T_h z_h / ||T_h z_h||. K_h depends only on the lag and on fixed parameters, T_h is fixed, and e is fixed. The composition is therefore a function of the multiset of (flat symbol id, position) pairs plus q, i.e. of v with the refinement s removed. Under arbitrary filler spacing the positions of the pair tokens are determined by filler counts independent of the role assignment, so F is constant on the fibre: no choice of H_L, state order or (rho, theta, w) changes this, and A(a) applies. To encode roles in the kernel one makes the recurrence coefficients functions of x_t, hence time-varying. For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, P6 gives h(s,s)=c_s, h(s+1,s)=a_(s+1) c_s, h(s+2,s)=(a_(s+2)a_(s+1)+b_(s+2)) c_s. Toeplitz forces c_s, a_s and b_s constant by induction on the lag, so (rho_t, theta_t) are constant. Non-Toeplitz h(t,s) has no single H(z) and the single-FFT path fails. This negates THAT path within M, not every conceivable O(L log L) algorithm. A(c) is therefore a conditional statement: IF duality is preserved, the bits must enter on the input side. QED.

A.3 Theorem A(d), A(e), A(f) and P2

A(d). By the Section 2 definition B = sum_t log2 |pi^(-1)(v_t)|, so an alphabet partition with |pi^(-1)| = 2 contributes exactly one bit per non-empty token and is COUNTED in B. An explicit tag and a symbol id carrying a role are the same quantity in a different carrier. Tagging only the KEY token of each of n pairs gives B = n, which attains B >= n from A(b). A positional prior is a support restriction: it reduces |A(v)| rather than increasing B, and under random filler spacing the restriction no longer holds, so the effective information falls to zero (measured 0.3438, 5 seeds). QED.

P2 (explicit construction). Fix keys and q; A = sum_p L_p v_p + c with d x d blocks L_p, read-out f(A,q) arbitrary, target j = j(q). Case 1: L_j singular. Pick nonzero delta in ker L_j and change only v_j; delta-A = 0, answer changes. Case 2: L_j invertible and some L_p != 0 for p != j. Pick delta_p and set delta_j = -L_j^(-1) L_p delta_p; then delta-A = L_p delta_p + L_j delta_j = 0, and delta_j is nonzero, so the target value changes. Case 3: L_j invertible and all other L_p = 0. Then A depends only on v_j; fix all values and redirect q to a different pair p != j, which changes the required answer while A is unchanged. In every case two legal assignments give the same (A,q) and different answers, so f cannot be exact. No step infers a target-coordinate perturbation from dim ker L > 0 alone. Scope: single mixer with a fixed linear historical summary; deep stacks with per-position nonlinear FFNs, input-dependent gates and outer-product states leave the premise. QED.

A(e). B < log2|A| makes exactness impossible for EVERY deterministic model by A(b), hence for every H_L; for B >= log2|A| Result 2(ii), under its stated premises, gives minimum depth H. The frontier is a right angle. A(f): under A4 the decoded value at depth H_L is k_(H_L); for H_L > H and distinct keys this differs from k_H. Retaining and selecting the H-th intermediate output gives the correct answer for any H_L >= H; the lower bound H_L >= H is unchanged. QED.

A.4 Result 2: induction and tightness

(i) P2 rules out an order-1 vector state with a linear summary; Construction C gives an order-2 tensor state with a contraction read-out, so the minimum is exactly 2. (ii) Induction on H. Base H=1: one contraction resolves one content match and yields the value; a zero-th layer cannot. Step: suppose H-1 layers resolve H-1 hops and produce q_H = C[k_H] by the unique H6 interface. Layer H performs one further match with q_H and returns k_H's successor. Fewer than H layers leave at least one hop unresolved: the last read-out sees a mixture of at least two successors and cannot be exact on all fibres. The analytic construction attains 1.0000 on the diagonal of the 5 x 5 grid and <= 0.10 off it, so the bound is tight and non-vacuous. Overshoot H_L > H is A(f), not a failure of the lower bound. QED.

A.5 Proposition P5: conditional state-order lower bound

Premise. The interface is H6 with rank at most D; the codebook rows are mutually orthogonal; at each of H hops there are c mutually orthogonal candidate successors; answers are determined by the joint choice of the start key and the H successors, so the candidate set has size c^(H+1). Claim. An order-o tensor state whose per-layer linear summary has image dimension at most o*d cannot be exact unless o >= H+1. Proof. The H+1 independent matching directions must be simultaneously recoverable from the summary for every one of the c^(H+1) joint choices. If o <= H, the summary is a linear map from a space of dimension (H+1)d into one of dimension o*D <= H*D; its kernel has dimension at least d > 0. By the P2 construction (with the blocks taken to be the coefficient matrices of the H+1 directions) there exist two distinct joint choices producing the same summary and the same q, hence the same read-out, yet requiring different answers. Exactness is impossible; therefore o >= H+1. This is a CONDITIONAL bound: when each hop offers only 2 candidates the same counting gives 2^(H+1) classes and the necessary order is weaker, degenerating to a log2(H+1) bit requirement. We state that degeneration explicitly rather than presenting c^(H+1) as unconditional. QED.

A.6 P1, P3, P4, P7, P8 and non-vacuity

P1. H(z) = sin(theta) / (1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)) gives y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t, a real-field equivalence; different rounding orders in the FFT and IIR paths make fp32 bitwise identity false (W1). P3. For a finite output set F with |F| <= |S|, the state can encode f(x) directly, so the literal claim 'O(1) states must lose' is false; the correct statement is that a SPECTRAL compression state loses content addressing. P4. At rho=1 and a single pole the Construction C state reduces to sum_tau k_tau v_tau^T, the linear-attention state; rho<1 retains

a decay spectrum. P7. With impulse response bounded by rho^lag / |sin(theta)| and per-step injection bounded by delta_max, the output error is at most delta_max / ((1-rho)|sin(theta)|), under a uniform floating-point perturbation model. P8. As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become collinear, so the grid is asymptotically rank deficient. Non-vacuity: n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing give |A|=4 at B=0 with acc <= 1/4, and exactness at B=2.

A.7 SE1: orthogonal-registration zero-crosstalk identity

PREMISE. C in R^(V x V) satisfies C C^T = I; S = sum_(a=1)^m w_a C[k_a]^T C[v_a] with distinct k_a; the query is q = C[k]. CLAIM. C[k] S = w_j C[V_K + v_j] when k = k_j, and 0 when k is not among the k_a. PROOF. C[k] C[k_a]^T = delta_(k,k_a) by row orthogonality, so C[k] S = sum_a w_a delta_(k,k_a) C[V_K+v_a]. Distinctness leaves at most one nonzero term. CONSEQUENCE. The read-out logits = (C[k] S) C[V_K:,:]^T are a one-hot vector scaled by w_j for a registered key, so accuracy is IDENTICALLY the fraction of correctly registered bindings. There is no cross-talk term to bound or estimate. NON-VACUITY. V = 64, V_K = 32, C the normalized Sylvester matrix, m >= 2: 200 randomized instances give a maximum on-target error of 4.4e-16 and maximum off-target error of 8.5e-17. BOUNDARY. Repeated keys, a rank-deficient codebook, or a non-codebook registration vector break the identity; this is why the break-alignment arm collapses to chance.

A.8 SE2: monotone improvement and cover time

PREMISE. n bindings arrive by i.i.d. sampling with replacement (coupon collector); the gate has false-accept at most alpha and miss at most beta; accepted experiences are correct; rho < 1 with a precision floor epsilon such that SE5's capacity K* >= n. CLAIM. (i) the number m_t of correct registrations is monotone non-decreasing; (ii) E[m_T] >= (1-beta) T until saturation; (iii) E[T_cover] = n H_n / (1-beta) with H_n = sum_(i=1)^n 1/i; (iv) Pr(T_cover > (1+eps) E T) <= exp(-eps^2 (1-beta) n H_n / 3); (v) the probability of any false commit in T rounds is at most min(1, T alpha). PROOF. (i) SE1 makes each correct registration a disjoint orthogonal term; register() replaces only the term of the same key and never removes another key's correct term, and K* >= n prevents decay below epsilon. (ii) The gate accepts a true binding with probability at least 1-beta. (iii) The classical coupon-collector expectation n H_n divided by the effective arrival probability 1-beta. (iv) Chernoff on a sum of n geometric variables, equivalently the union bound Pr(some type missing) <= n(1-1/n)^T. (v) Boole's inequality over T rounds. NON-VACUITY AND CHECK. For n = 32, alpha = 0.05, beta approximately 0, the prediction is 129.87 rounds; the measured five-seed mean first full-cover round is 130.00 (relative gap 0.10 percent), with per-seed values 170, 100, 90, 120, 170. With n = 5 the sample SD is 38.08 and the 95% t-CI is [82.72, 177.28] rounds: the prediction lies inside the interval, but the interval spans 73% of the mean, so this is weak evidence and the small gap is NOT a precision claim. The arrival model was set to i.i.d. BEFORE measuring. BOUNDARY. A deterministic round-robin schedule gives T_cover = n and SE2 does not apply; large beta stretches the cover time by 1/(1-beta); if K* < n then SE5 takes over and full cover is impossible.

A.9 SE3: the statistical gate is necessary

PREMISE. Each accepted experience is corrupted independently with probability p; a corrupted value is uniform on [0, V_K); there is no gate; the final value of a key is its last write. CLAIM. E[acc] = (1-p) + p/V_K = 1 - p(1 - 1/V_K) exactly, and E[acc] <= that value in general because some keys may never arrive. PROOF. The last write is correct either when it is not corrupted (probability 1-p) or when corruption happens to draw the true value (probability p/V_K). SE1 makes the keys independent, so accuracy is the mean of n such Bernoulli variables. CONCENTRATION. Over N independent key-level trials, Hoeffding gives Pr(|a_bar - mu| >= t) <= 2 exp(-2 N t^2). With N = 640000 the 4-sigma interval is [0.7071, 0.7116] and the enumeration mean is 0.7091, inside it. WHY NECESSARY. Orthogonality prevents cross-talk but cannot prevent a wrong value from occupying an orthogonal slot; hence without a gate SE2's full cover is unattainable under corruption. SCOPE. With p = 0, mu = 1 and the gate is irrelevant: the ungated clean arm ties the fused arm exactly. The gate is necessary ONLY when experiences can be wrong. W15. An earlier draft called mu an almost-sure upper bound and tested it pointwise against a sample mean; a sample mean legitimately exceeds an expectation. The claim is now an exact expectation with a concentration interval.

A.10 SE4: compatibility with the CT7 rate bound

CT7(b) bounds the information that T consolidation rounds can write into the state by T w b bits, independent of context length L and of bank capacity C. SE2 permits about (1-beta) T registrations and a corresponding accuracy gain. CLAIM. Both hold together. PROOF. Registration is injective (SE1's partial isometry), so one binding corresponds to one key-value association of exactly log2(V_K) bits. With T = 100 and w = 1, SE2 registers 100 bindings using 500 bits, while CT7 allows 500 bits: equality, not violation. MEANING. Self-evolution is real but bounded. Long context buys inference-time retrieval, not a faster learning rate. CT7(c)'s earlier statement that improvement was not proved is resolved for this finite process; CT7(b)'s upper bound is untouched.

A.11 SE5: optimal decay eviction and closed-form capacity

PREMISE. Slot weights w_a = rho^age(a) with rho < 1; utility is additive and orthogonal (SE1); precision floor epsilon; capacity C_max. CLAIM. (i) Greedy removal of the smallest weight is optimal; (ii) the number of slots whose weight stays above epsilon is K* = floor(log epsilon / log rho). PROOF. (i) Utility U = sum_a w_a u_a with u_a >= 0 and mutually independent; retaining the C_max largest weights maximizes U, and dropping the smallest is its online realization. (ii) rho^a > epsilon iff a < log epsilon / log rho. NON-VACUITY. At rho = 0.999 and epsilon = 1e-3, K* = 6904; at rho = 0.90, K* = 65. The fast-decay arm loses accuracy to 0.1313 while the standard arm stays at 1.0000 over 320 rounds, matching the predicted capacity collapse. BOUNDARY. rho = 1 is critically stable, non-BIBO and has amplitude growing with L; the architecture excludes it and no run instantiates it. A near-critical value such as 0.99999 barely decays within the run and therefore has no discriminating power; 0.90 was chosen so SE5 is observable.

Appendix B Withdrawn Statements

ID	Withdrawn statement and reason
W1	Unconditional fp32 bitwise FFT/IIR duality: different rounding orders; real-field equivalence only.
W8	Content addressing requires input-dependent kernel coefficients: refuted by Construction C with strictly static kernels.
W9	Absolute-position parity is a general role basis: a hidden format prior; collapses under arbitrary spacing.
W10/W11	'Exactly one bit per token is necessary and sufficient': the necessary quantity is total B >= log2|A|; one bit per token is a sufficient carrier.
W12	Single-hop attention parity (Fisher p=0.4998): the stored baseline did not reproduce in the same-run audit (Section 4.6). No equivalence or superiority claim is made.
W13	Trained corroboration of the layer bound under the earlier protocol: withdrawn after the fairness audit removed two input asymmetries. Re-established on the corrected architecture in Section 4.12; the earlier number is not reused.
W14	Straight-through inter-layer normalisation (norm taken outside the graph): finite-difference gradient deviation 2.5e-03. Replaced by a differentiable square root built from log and exp, deviation 2.9e-10; all conservation checks are rerun on final weights.
W15	SE3 as an almost-sure upper bound tested pointwise against a sample mean: a finite sample mean legitimately exceeds an expectation. Replaced by the exact expectation 1-p(1-1/V_K) with a Hoeffding concentration interval (Appendix A.9).

Appendix C Reproducibility

The submission is accompanied by an anonymised artifact containing the reference implementation of the layer, the seven same-run mechanism references used only for the Section 4.6 audit, the reachability enumeration of Section 4.3, and all raw result files. Every number in Tables 2-4 is injected from those files at build time; no value is transcribed by hand. All probes are CPU-only NumPy and complete in minutes. The Section 4.12 run uses 1500 Adam steps, lr 0.02, batch 64, five seeds, and 512 training plus 512 held-out instances per seed; the architecture module and every

convergence curve are in the artifact. The fusion run of Section 4.13 and Appendices D-E uses 320 i.i.d. arrival rounds, 32 bindings, five seeds, rho = 0.999 and epsilon = 1e-3; all seven arms, sixteen ablation cells and every individual seed are printed in Appendix D. Seeds are listed per table. The environment, dependency versions and a one-command reproduction script are in the artifact. The 2027 call for papers and submission deadline were not published at the time of writing; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

No count of statements is presented as a contribution. The withdrawn-statement register in Appendix B is a soundness practice, not an originality claim.

Appendix D All Raw Fusion Measurements

Every row below is a single seed of a single arm. Nothing is averaged away and no value is transcribed by hand: the generator reads the raw result files at build time. Protocol: 32 bindings, 320 rounds of i.i.d. arrival, rho = 0.999 unless stated, precision floor epsilon = 1e-3, capacity 64, one-sided paired-CI gate with alpha = 0.05, tau = 0, 8 probes and noise sigma = 0.10. Chance is 1/32 = 0.0313. The core state is 32768 bytes in every row (d = 64, fp64; notation table in Section 2). A machine-readable long-form CSV of every row below is supplied in the artifact as appendix_D_fusion_long.csv; no value is OCR-transcribed.

arm	seed	final acc	commits	first full cover
fused gated	0	1.0000	49	170
fused gated	1	1.0000	55	100
fused gated	2	1.0000	47	90
fused gated	3	1.0000	53	120
fused gated	4	1.0000	54	170
no consolidation	0	0.0000	0	none
no consolidation	1	0.0000	0	none
no consolidation	2	0.0000	0	none
no consolidation	3	0.0000	0	none
no consolidation	4	0.0000	0	none
ungated clean	0	1.0000	320	170
ungated clean	1	1.0000	320	100
ungated clean	2	1.0000	320	90
ungated clean	3	1.0000	320	120
ungated clean	4	1.0000	320	170
ungated corrupt	0	0.6250	320	none
ungated corrupt	1	0.7500	320	none
ungated corrupt	2	0.5625	320	none
ungated corrupt	3	0.5625	320	none
ungated corrupt	4	0.7188	320	none
random write	0	0.6250	154	none
random write	1	0.6875	153	none
random write	2	0.5938	154	none
random write	3	0.7500	155	none
random write	4	0.7812	155	none
break alignment	0	0.0000	49	none
break alignment	1	0.0000	55	none
break alignment	2	0.0938	47	none
break alignment	3	0.0000	53	none
break alignment	4	0.0312	54	none
rho fast	0	0.1562	49	none
rho fast	1	0.1562	55	none
rho fast	2	0.0938	47	none
rho fast	3	0.1250	53	none
rho fast	4	0.1250	54	none

Table 18. All 35 arm-seed runs. The first-full-cover column reports the first sampled round at which accuracy reaches 1.0 (sampling every 10 rounds); '-' means no such round occurred. The five fused values are the ones averaged in Section 4.13.

corruption	config	mean	sd	n
0.00	gate off	1.0000	0.0000	5
0.00	alpha 0.10	1.0000	0.0000	5
0.00	alpha 0.05	1.0000	0.0000	5
0.00	alpha 0.01	1.0000	0.0000	5
0.00	tau positive	1.0000	0.0000	5
0.00	sigma 0.30	1.0000	0.0000	5
0.00	rho 0.90	0.1313	0.0234	5
0.00	rho 0.999	1.0000	0.0000	5
0.30	gate off	0.6438	0.0781	5
0.30	alpha 0.10	1.0000	0.0000	5
0.30	alpha 0.05	1.0000	0.0000	5
0.30	alpha 0.01	1.0000	0.0000	5
0.30	tau positive	1.0000	0.0000	5
0.30	sigma 0.30	1.0000	0.0000	5
0.30	rho 0.90	0.1000	0.0364	5
0.30	rho 0.999	1.0000	0.0000	5

Table 19. All 16 corruption x configuration cells, each over five seeds. Under zero corruption the six gate configurations saturate and only the fast-decay configuration degrades; under 30 percent corruption the gate-off row degrades while every gated row saturates. This is the measured form of SE3's scope.

check	pass
theory 7of7	yes
theory SE1	yes
theory SE2	yes
theory SE3	yes
theory SE4	yes
theory SE5	yes
theory nonvacuous SE2	yes
theory SE3 concentration	yes
theory SE3 exact expectation	yes
evolution all invariants	yes
evolution monotone	yes
fused above 0.90	yes
all key paired positive	yes
core state bytes constant across rounds	yes
slot count bounded by capacity	yes
SE5 capacity predicts rho fast loss	yes
ablation clean gate configs saturate	yes
ablation noisy gate necessary	yes
ablation rho decay matches SE5	yes
SE2 prediction matches measurement	yes
SE3 concentration holds	yes
SE3 enumeration within 4sigma	yes
no gradient through retrieval	yes
no NxN score	yes
no kv cache	yes
v64 stack invariants	yes
v64 interface identities	yes

Table 20. All 27 consistency checks on the FINAL post-evolution state. They cover the five analytic results, the paired statistics, monotonicity, core-state constancy, capacity bounds, the SE2 prediction-measurement agreement, SE3's concentration interval, the v6.4 stack invariants and the interface identities. A false here would be a finding, not a crashed run.

Appendix E Premise, Boundary and Non-Vacuity Ledger

A theorem whose premises admit no instance is empty. The table below gives one concrete satisfying instance, the exact boundary at which the statement stops, and what would falsify it, for every new supporting result. These are not additional contributions; they are the audit trail that keeps the premises honest.

result	satisfying instance	boundary / falsifier
SE1	V=64, V_K=32, Sylvester C, m>=2 distinct keys	repeated keys or rank-deficient code; any nonzero off-target read
SE2	n=32 i.i.d. arrivals, beta~0, K*=6904 >= n	round-robin arrival; beta>0; K*<n; a non-monotone fused curve
SE3	p=0.30, uniform corrupted value, no gate	p=0 makes the gate irrelevant; a gated arm below the exact expectation
SE4	T=100, w=1, b=log2(32)=5, 500 bits used = 500 allowed	any run writing more than T w log2(V_K) bits into the state
SE5	rho=0.999, eps=1e-3, K*=6904; rho=0.90, K*=65	rho=1 is non-BIBO and excluded; a better-than-greedy eviction
CT7	answer() is no-grad; selection Jacobian vanishes a.e.	a differentiable exact-retrieval channel through the online path

Table 21. Each row names a measurement that would have to come out differently. The five supporting results SE1-SE5 do not change the four core results of Section 1.1; they make the empirical claim in Result 4 auditable at the level of individual runs. The withdrawn-statement register (Appendix B) records W13, W14 and W15, all of which are premise corrections rather than numerical adjustments.

Appendix F All 1500-Step Convergence Curves

Section 4.12 reports only the diagonal and the first-crossing step. Here every sampled point of every arm is printed for all five seeds, followed by every individual cell of the layer-by-hop matrix. Protocol: 1500 Adam steps, lr 0.02, batch 64, 512 training and 512 held-out instances per seed, width d = 64, fixed Hadamard embedding, evaluation every 150 steps. The architecture module's inter-layer map is the fixed codebook value-to-key swap; only the query projection and read-out are learned, from random

initialisation.

arm	step	mean acc	sd	n seeds
arch codebook M	0	0.0266	0.0064	5
arch codebook M	150	1.0000	0.0000	5
arch codebook M	300	1.0000	0.0000	5
arch codebook M	450	1.0000	0.0000	5
arch codebook M	600	1.0000	0.0000	5
arch codebook M	750	1.0000	0.0000	5
arch codebook M	900	1.0000	0.0000	5
arch codebook M	1050	1.0000	0.0000	5
arch codebook M	1200	1.0000	0.0000	5
arch codebook M	1350	1.0000	0.0000	5
arch codebook M	1500	1.0000	0.0000	5
arch analytic witness	0	1.0000	0.0000	5
legacy learnable interface random	0	0.0301	0.0052	5
legacy learnable interface random	150	0.2812	0.0368	5
legacy learnable interface random	300	0.2633	0.0381	5
legacy learnable interface random	450	0.2492	0.0319	5
legacy learnable interface random	600	0.2398	0.0278	5
legacy learnable interface random	750	0.2383	0.0388	5
legacy learnable interface random	900	0.2422	0.0276	5
legacy learnable interface random	1050	0.2445	0.0321	5
legacy learnable interface random	1200	0.2520	0.0329	5
legacy learnable interface random	1350	0.2535	0.0311	5
legacy learnable interface random	1500	0.2492	0.0325	5
legacy learnable interface fixedM	0	0.0387	0.0119	5
legacy learnable interface fixedM	150	1.0000	0.0000	5
legacy learnable interface fixedM	300	1.0000	0.0000	5
legacy learnable interface fixedM	450	1.0000	0.0000	5
legacy learnable interface fixedM	600	1.0000	0.0000	5
legacy learnable interface fixedM	750	1.0000	0.0000	5
legacy learnable interface fixedM	900	1.0000	0.0000	5
legacy learnable interface fixedM	1050	1.0000	0.0000	5
legacy learnable interface fixedM	1200	1.0000	0.0000	5
legacy learnable interface fixedM	1350	1.0000	0.0000	5
legacy learnable interface fixedM	1500	1.0000	0.0000	5
legacy trainable near-swap	0	0.0238	0.0106	5
legacy trainable near-swap	150	1.0000	0.0000	5
legacy trainable near-swap	300	1.0000	0.0000	5
legacy trainable near-swap	450	1.0000	0.0000	5
legacy trainable near-swap	600	1.0000	0.0000	5
legacy trainable near-swap	750	1.0000	0.0000	5
legacy trainable near-swap	900	1.0000	0.0000	5
legacy trainable near-swap	1050	1.0000	0.0000	5
legacy trainable near-swap	1200	1.0000	0.0000	5
legacy trainable near-swap	1350	1.0000	0.0000	5
legacy trainable near-swap	1500	1.0000	0.0000	5

Table 22. All 45 arm-step cells. The random learnable interface stays in 0.21-0.29 for the entire budget (this task's chance is 1/32 = 0.0313; 0.25 belongs to a DIFFERENT historical task); the codebook-fixed interface and the near-swap basin cross 0.99 by step 150 on every seed. The analytic witness is exact at step 0 by construction.

hops	layers	seed	acc	params	steps
1	1	0	1.0000	6144	1500
1	1	1	1.0000	6144	1500
1	1	2	1.0000	6144	1500
1	1	3	1.0000	6144	1500
1	1	4	1.0000	6144	1500
1	2	0	0.0215	6144	1500
1	2	1	0.0430	6144	1500
1	2	2	0.0332	6144	1500
1	2	3	0.0312	6144	1500
1	2	4	0.0273	6144	1500
1	3	0	0.0352	6144	1500
1	3	1	0.0410	6144	1500
1	3	2	0.0312	6144	1500
1	3	3	0.0312	6144	1500
1	3	4	0.0566	6144	1500
1	4	0	0.0469	6144	1500
1	4	1	0.0410	6144	1500
1	4	2	0.0352	6144	1500
1	4	3	0.0312	6144	1500
1	4	4	0.0273	6144	1500
2	1	0	0.1582	6144	1500
2	1	1	0.1543	6144	1500
2	1	2	0.2129	6144	1500
2	1	3	0.1875	6144	1500
2	1	4	0.1914	6144	1500
2	2	0	1.0000	6144	1500
2	2	1	1.0000	6144	1500
2	2	2	1.0000	6144	1500
2	2	3	1.0000	6144	1500
2	2	4	1.0000	6144	1500
2	3	0	0.0352	6144	1500
2	3	1	0.0254	6144	1500
2	3	2	0.0273	6144	1500
2	3	3	0.0312	6144	1500
2	3	4	0.0312	6144	1500
2	4	0	0.0352	6144	1500
2	4	1	0.0410	6144	1500
2	4	2	0.0371	6144	1500
2	4	3	0.0273	6144	1500
2	4	4	0.0332	6144	1500
3	1	0	0.1562	6144	1500
3	1	1	0.1660	6144	1500
3	1	2	0.2051	6144	1500
3	1	3	0.1230	6144	1500
3	1	4	0.1348	6144	1500
3	2	0	0.2363	6144	1500
3	2	1	0.2480	6144	1500
3	2	2	0.2500	6144	1500
3	2	3	0.2773	6144	1500
3	2	4	0.2246	6144	1500
3	3	0	1.0000	6144	1500
3	3	1	1.0000	6144	1500
3	3	2	1.0000	6144	1500
3	3	3	1.0000	6144	1500
3	3	4	1.0000	6144	1500
3	4	0	0.0352	6144	1500
3	4	1	0.0449	6144	1500
3	4	2	0.0332	6144	1500
3	4	3	0.0352	6144	1500
3	4	4	0.0332	6144	1500
4	1	0	0.1348	6144	1500
4	1	1	0.1875	6144	1500
4	1	2	0.1719	6144	1500
4	1	3	0.1504	6144	1500
4	1	4	0.1309	6144	1500
4	2	0	0.2402	6144	1500
4	2	1	0.2051	6144	1500
4	2	2	0.2637	6144	1500
4	2	3	0.2773	6144	1500
4	2	4	0.2207	6144	1500
4	3	0	0.3555	6144	1500
4	3	1	0.3516	6144	1500
4	3	2	0.3496	6144	1500
4	3	3	0.3633	6144	1500
4	3	4	0.3535	6144	1500
4	4	0	1.0000	6144	1500
4	4	1	1.0000	6144	1500
4	4	2	1.0000	6144	1500
4	4	3	1.0000	6144	1500
4	4	4	1.0000	6144	1500

Table 23. All 80 layer-by-hop runs. Parameter count is 6144 in every row because the inter-layer map is shared and is not a parameter; depth adds computation, not weights. The diagonal is the only set of cells above 0.99.

Appendix G All Fusion Evolution Curves

Section 4.13 reports final accuracies. Here the full trajectory of each arm is printed: accuracy is sampled every 10 rounds and averaged over the five seeds at each sampled round. Protocol: 32 bindings, 320 i.i.d. arrival rounds, rho = 0.999 except the fast-decay arm, epsilon = 1e-3, capacity 64, one-sided paired-CI gate. The core state is 32768 bytes at every sampled point of every run.

arm	round	mean acc	sd	mean slots
fused gated	1	0.0312	0.0000	1.0
fused gated	10	0.2687	0.0250	8.6
fused gated	20	0.4688	0.0198	15.0
fused gated	30	0.6562	0.0685	21.0
fused gated	40	0.7438	0.0459	23.8
fused gated	50	0.7812	0.0442	25.0
fused gated	60	0.8562	0.0153	27.4
fused gated	70	0.9000	0.0415	28.8
fused gated	80	0.9125	0.0415	29.2
fused gated	90	0.9500	0.0375	30.4
fused gated	100	0.9625	0.0364	30.8
fused gated	110	0.9750	0.0234	31.2
fused gated	120	0.9812	0.0250	31.4
fused gated	130	0.9812	0.0250	31.4
fused gated	140	0.9875	0.0153	31.6
fused gated	150	0.9875	0.0153	31.6
fused gated	160	0.9875	0.0153	31.6
fused gated	170	1.0000	0.0000	32.0
fused gated	180	1.0000	0.0000	32.0
fused gated	190	1.0000	0.0000	32.0
fused gated	200	1.0000	0.0000	32.0
fused gated	210	1.0000	0.0000	32.0
fused gated	220	1.0000	0.0000	32.0

fused gated	230	1.0000	0.0000	32.0
fused gated	240	1.0000	0.0000	32.0
fused gated	250	1.0000	0.0000	32.0
fused gated	260	1.0000	0.0000	32.0
fused gated	270	1.0000	0.0000	32.0
fused gated	280	1.0000	0.0000	32.0
fused gated	290	1.0000	0.0000	32.0
fused gated	300	1.0000	0.0000	32.0
fused gated	310	1.0000	0.0000	32.0
fused gated	320	1.0000	0.0000	32.0
no consolidation	1	0.0000	0.0000	0.0
no consolidation	10	0.0000	0.0000	0.0
no consolidation	20	0.0000	0.0000	0.0
no consolidation	30	0.0000	0.0000	0.0
no consolidation	40	0.0000	0.0000	0.0
no consolidation	50	0.0000	0.0000	0.0
no consolidation	60	0.0000	0.0000	0.0
no consolidation	70	0.0000	0.0000	0.0
no consolidation	80	0.0000	0.0000	0.0
no consolidation	90	0.0000	0.0000	0.0
no consolidation	100	0.0000	0.0000	0.0
no consolidation	110	0.0000	0.0000	0.0
no consolidation	120	0.0000	0.0000	0.0
no consolidation	130	0.0000	0.0000	0.0
no consolidation	140	0.0000	0.0000	0.0
no consolidation	150	0.0000	0.0000	0.0
no consolidation	160	0.0000	0.0000	0.0
no consolidation	170	0.0000	0.0000	0.0
no consolidation	180	0.0000	0.0000	0.0
no consolidation	190	0.0000	0.0000	0.0
no consolidation	200	0.0000	0.0000	0.0
no consolidation	210	0.0000	0.0000	0.0
no consolidation	220	0.0000	0.0000	0.0
no consolidation	230	0.0000	0.0000	0.0
no consolidation	240	0.0000	0.0000	0.0
no consolidation	250	0.0000	0.0000	0.0
no consolidation	260	0.0000	0.0000	0.0
no consolidation	270	0.0000	0.0000	0.0
no consolidation	280	0.0000	0.0000	0.0
no consolidation	290	0.0000	0.0000	0.0
no consolidation	300	0.0000	0.0000	0.0
no consolidation	310	0.0000	0.0000	0.0
no consolidation	320	0.0000	0.0000	0.0
ungated clean	1	0.0312	0.0000	1.0
ungated clean	10	0.2687	0.0250	8.6
ungated clean	20	0.4688	0.0198	15.0
ungated clean	30	0.6562	0.0685	21.0
ungated clean	40	0.7438	0.0459	23.8
ungated clean	50	0.7812	0.0442	25.0
ungated clean	60	0.8562	0.0153	27.4
ungated clean	70	0.9000	0.0415	28.8
ungated clean	80	0.9125	0.0415	29.2
ungated clean	90	0.9500	0.0375	30.4
ungated clean	100	0.9625	0.0364	30.8
ungated clean	110	0.9750	0.0234	31.2
ungated clean	120	0.9812	0.0250	31.4
ungated clean	130	0.9812	0.0250	31.4
ungated clean	140	0.9875	0.0153	31.6
ungated clean	150	0.9875	0.0153	31.6
ungated clean	160	0.9875	0.0153	31.6
ungated clean	170	1.0000	0.0000	32.0
ungated clean	180	1.0000	0.0000	32.0
ungated clean	190	1.0000	0.0000	32.0
ungated clean	200	1.0000	0.0000	32.0
ungated clean	210	1.0000	0.0000	32.0
ungated clean	220	1.0000	0.0000	32.0
ungated clean	230	1.0000	0.0000	32.0
ungated clean	240	1.0000	0.0000	32.0
ungated clean	250	1.0000	0.0000	32.0
ungated clean	260	1.0000	0.0000	32.0
ungated clean	270	1.0000	0.0000	32.0
ungated clean	280	1.0000	0.0000	32.0
ungated clean	290	1.0000	0.0000	32.0
ungated clean	300	1.0000	0.0000	32.0
ungated clean	310	1.0000	0.0000	32.0
ungated clean	320	1.0000	0.0000	32.0
ungated corrupt	1	0.0125	0.0153	1.0
ungated corrupt	10	0.1750	0.0153	8.6
ungated corrupt	20	0.3125	0.0713	15.2
ungated corrupt	30	0.4250	0.0580	20.4
ungated corrupt	40	0.4875	0.0424	23.4
ungated corrupt	50	0.5813	0.0580	25.6
ungated corrupt	60	0.6188	0.0893	26.8
ungated corrupt	70	0.6188	0.0459	28.4
ungated corrupt	80	0.6375	0.0424	29.8
ungated corrupt	90	0.6750	0.0319	30.2
ungated corrupt	100	0.6813	0.0306	31.0
ungated corrupt	110	0.6750	0.0424	31.2
ungated corrupt	120	0.6937	0.0234	31.4
ungated corrupt	130	0.6750	0.0468	31.4
ungated corrupt	140	0.6937	0.0606	31.6
ungated corrupt	150	0.7125	0.0500	32.0
ungated corrupt	160	0.7500	0.0559	32.0
ungated corrupt	170	0.7125	0.0800	32.0
ungated corrupt	180	0.7125	0.0824	32.0
ungated corrupt	190	0.7188	0.1296	32.0
ungated corrupt	200	0.7188	0.1479	32.0
ungated corrupt	210	0.7562	0.1389	32.0
ungated corrupt	220	0.7438	0.0956	32.0
ungated corrupt	230	0.7312	0.1019	32.0
ungated corrupt	240	0.7500	0.0625	32.0
ungated corrupt	250	0.6687	0.0673	32.0
ungated corrupt	260	0.6687	0.0755	32.0
ungated corrupt	270	0.6937	0.0415	32.0
ungated corrupt	280	0.7188	0.0713	32.0
ungated corrupt	290	0.7063	0.0153	32.0
ungated corrupt	300	0.6562	0.0198	32.0
ungated corrupt	310	0.6312	0.0234	32.0
ungated corrupt	320	0.6438	0.0781	32.0
random write	1	0.0000	0.0000	0.2
random write	10	0.1000	0.0125	4.8
random write	20	0.1250	0.0198	6.0
random write	30	0.1625	0.0459	7.8
random write	40	0.2500	0.0198	11.2
random write	50	0.3375	0.0125	15.0
random write	60	0.4062	0.0685	17.4
random write	70	0.4562	0.0805	20.6
random write	80	0.5062	0.0538	22.2
random write	90	0.5188	0.0468	22.6
random write	100	0.5250	0.0667	24.2
random write	110	0.5437	0.0545	25.0
random write	120	0.5750	0.0319	25.4
random write	130	0.5625	0.0625	26.0
random write	140	0.5750	0.0643	26.8
random write	150	0.6000	0.0637	27.0
random write	160	0.6375	0.1038	28.0
random write	170	0.6625	0.0871	28.8
random write	180	0.6750	0.0755	29.2
random write	190	0.7125	0.0935	29.4
random write	200	0.7000	0.1179	29.8
random write	210	0.7375	0.1000	30.0
random write	220	0.7250	0.0956	30.4
random write	230	0.7375	0.0897	30.8
random write	240	0.7500	0.0765	31.0
random write	250	0.6875	0.1046	31.2
random write	260	0.6875	0.1064	31.2
random write	270	0.7188	0.0948	31.4
random write	280	0.7063	0.0919	31.4
random write	290	0.7000	0.0805	31.4
random write	300	0.6813	0.0696	31.4
random write	310	0.6687	0.0508	31.4
random write	320	0.6875	0.0713	31.4
break alignment	1	0.0437	0.0250	1.0
break alignment	10	0.0312	0.0342	8.6
break alignment	20	0.0250	0.0125	15.0
break alignment	30	0.0312	0.0342	21.0
break alignment	40	0.0187	0.0250	23.8
break alignment	50	0.0187	0.0250	25.0
break alignment	60	0.0312	0.0198	27.4
break alignment	70	0.0312	0.0342	28.8
break alignment	80	0.0312	0.0342	29.2
break alignment	90	0.0312	0.0342	30.4
break alignment	100	0.0312	0.0395	30.8
break alignment	110	0.0125	0.0250	31.2
break alignment	120	0.0125	0.0250	31.4
break alignment	130	0.0063	0.0125	31.4
break alignment	140	0.0187	0.0153	31.6
break alignment	150	0.0125	0.0153	31.6
break alignment	160	0.0125	0.0153	31.6
break alignment	170	0.0250	0.0234	32.0
break alignment	180	0.0250	0.0234	32.0
break alignment	190	0.0187	0.0153	32.0
break alignment	200	0.0125	0.0153	32.0
break alignment	210	0.0125	0.0153	32.0
break alignment	220	0.0187	0.0250	32.0
break alignment	230	0.0312	0.0280	32.0
break alignment	240	0.0312	0.0280	32.0
break alignment	250	0.0250	0.0234	32.0
break alignment	260	0.0187	0.0250	32.0
break alignment	270	0.0187	0.0153	32.0
break alignment	280	0.0187	0.0153	32.0
break alignment	290	0.0125	0.0153	32.0
break alignment	300	0.0250	0.0364	32.0
break alignment	310	0.0250	0.0364	32.0

break alignment	320	0.0250	0.0364	32.0	2	0	2	4	0.2500	0.0e+00
rho fast	1	0.0312	0.0000	1.0	2	0	3	4	0.2500	0.0e+00
rho fast	10	0.2687	0.0250	8.6	2	0	4	4	0.2500	0.0e+00
rho fast	20	0.4688	0.0198	15.0	2	0	5	4	0.2500	0.0e+00
rho fast	30	0.6562	0.0685	21.0	2	0	6	4	0.2500	0.0e+00
rho fast	40	0.7438	0.0459	23.8	2	0	7	4	0.2500	0.0e+00
rho fast	50	0.7812	0.0442	25.0	2	0	8	4	0.2500	0.0e+00
rho fast	60	0.8562	0.0153	27.4	2	1	1	4	0.5000	0.0e+00
rho fast	70	0.7625	0.0755	28.8	2	1	2	4	0.5000	0.0e+00
rho fast	80	0.5813	0.0580	29.2	2	1	3	4	0.5000	0.0e+00
rho fast	90	0.4188	0.0897	30.4	2	1	4	4	0.5000	0.0e+00
rho fast	100	0.3250	0.1000	30.8	2	1	5	4	0.5000	0.0e+00
rho fast	110	0.3063	0.0893	31.2	2	1	6	4	0.5000	0.0e+00
rho fast	120	0.2750	0.0824	31.4	2	1	7	4	0.5000	0.0e+00
rho fast	130	0.2375	0.0755	31.4	2	1	8	4	0.5000	0.0e+00
rho fast	140	0.2313	0.0643	31.6	2	2	1	4	1.0000	0.0e+00
rho fast	150	0.2000	0.0755	31.6	2	2	2	4	1.0000	0.0e+00
rho fast	160	0.1750	0.0545	31.6	2	2	3	4	1.0000	0.0e+00
rho fast	170	0.1562	0.0656	32.0	2	2	4	4	1.0000	0.0e+00
rho fast	180	0.1562	0.0442	32.0	2	2	5	4	1.0000	0.0e+00
rho fast	190	0.1875	0.0523	32.0	2	2	6	4	1.0000	0.0e+00
rho fast	200	0.1500	0.0750	32.0	2	2	7	4	1.0000	0.0e+00
rho fast	210	0.1500	0.0871	32.0	2	2	8	4	1.0000	0.0e+00
rho fast	220	0.1500	0.0696	32.0	3	0	1	8	0.1250	0.0e+00
rho fast	230	0.1250	0.0656	32.0	3	0	2	8	0.1250	0.0e+00
rho fast	240	0.1375	0.0643	32.0	3	0	3	8	0.1250	0.0e+00
rho fast	250	0.1187	0.0306	32.0	3	0	4	8	0.1250	0.0e+00
rho fast	260	0.1125	0.0250	32.0	3	0	5	8	0.1250	0.0e+00
rho fast	270	0.1375	0.0153	32.0	3	0	6	8	0.1250	0.0e+00
rho fast	280	0.1250	0.0280	32.0	3	0	7	8	0.1250	0.0e+00
rho fast	290	0.1313	0.0364	32.0	3	0	8	8	0.1250	0.0e+00
rho fast	300	0.1437	0.0250	32.0	3	1	1	8	0.2500	0.0e+00
rho fast	310	0.1250	0.0342	32.0	3	1	2	8	0.2500	0.0e+00
rho fast	320	0.1313	0.0234	32.0	3	1	3	8	0.2500	0.0e+00
					3	1	4	8	0.2500	0.0e+00
					3	1	5	8	0.2500	0.0e+00
					3	1	6	8	0.2500	0.0e+00
					3	1	7	8	0.2500	0.0e+00
					3	1	8	8	0.2500	0.0e+00
					3	3	1	8	1.0000	0.0e+00
					3	3	2	8	1.0000	0.0e+00
					3	3	3	8	1.0000	0.0e+00
					3	3	4	8	1.0000	0.0e+00
					3	3	5	8	1.0000	0.0e+00
					3	3	6	8	1.0000	0.0e+00
					3	3	7	8	1.0000	0.0e+00
					3	3	8	8	1.0000	0.0e+00
					4	0	1	16	0.0625	0.0e+00
					4	0	2	16	0.0625	0.0e+00
					4	0	3	16	0.0625	0.0e+00
					4	0	4	16	0.0625	0.0e+00
					4	0	5	16	0.0625	0.0e+00
					4	0	6	16	0.0625	0.0e+00
					4	0	7	16	0.0625	0.0e+00
					4	0	8	16	0.0625	0.0e+00
					4	1	1	16	0.1250	0.0e+00
					4	1	2	16	0.1250	0.0e+00
					4	1	3	16	0.1250	0.0e+00
					4	1	4	16	0.1250	0.0e+00
					4	1	5	16	0.1250	0.0e+00
					4	1	6	16	0.1250	0.0e+00
					4	1	7	16	0.1250	0.0e+00
					4	1	8	16	0.1250	0.0e+00
					4	4	1	16	1.0000	0.0e+00
					4	4	2	16	1.0000	0.0e+00
					4	4	3	16	1.0000	0.0e+00
					4	4	4	16	1.0000	0.0e+00
					4	4	5	16	1.0000	0.0e+00
					4	4	6	16	1.0000	0.0e+00
					4	4	7	16	1.0000	0.0e+00
					4	4	8	16	1.0000	0.0e+00
					1	1	1	2	1.0000	0.0e+00
					1	1	2	2	0.0000	0.0e+00
					1	1	3	2	0.0000	0.0e+00
					1	1	4	2	0.0000	0.0e+00
					1	1	5	2	0.0586	0.0e+00
					2	2	1	4	0.0000	0.0e+00
					2	2	2	4	1.0000	0.0e+00
					2	2	3	4	0.0000	0.0e+00
					2	2	4	4	0.0000	0.0e+00
					2	2	5	4	0.0625	0.0e+00
					3	3	1	8	0.0000	0.0e+00
					3	3	2	8	0.0000	0.0e+00
					3	3	3	8	1.0000	0.0e+00
					3	3	4	8	0.0000	0.0e+00
					3	3	5	8	0.0801	0.0e+00
					4	4	1	16	0.0000	0.0e+00

Table 24. All 231 arm-round cells. The fused arm is monotone on all five seeds; the ungated corrupted arm rises then fluctuates as later writes overwrite earlier correct registrations; the fast-decay arm loses slots as SE5's finite-precision capacity is exhausted.

comparison	paired diff	CI95 low	CI95 high	all pos
no consolidation	+1.0000	+1.0000	+1.0000	yes
ungated clean	+0.0000	+0.0000	+0.0000	no
ungated corrupt	+0.3563	+0.2875	+0.4250	yes
random write	+0.3125	+0.2500	+0.3750	yes
break alignment	+0.9750	+0.9375	+1.0000	yes
rho fast	+0.8688	+0.8500	+0.8875	yes

Table 25. All 6 paired bootstrap comparisons, 10000 resamples, five seeds. The ungated clean arm ties exactly: with no corruption the gate is irrelevant, which is the measured form of SE3's scope, not a failure of the comparison.

Appendix H Exhaustive Information-Bound Enumeration

Theorem A is checked against the complete enumeration of the binary-chain fibre space, not against a sample. Every (m, B) pair below is enumerated over all 2^m role assignments with bit-identical visible inputs. The depth scan then verifies that stacking 1-8 static-LTI layers produces bit-identical outputs across compatible role assignments. These are finite exhaustive objects; they certify non-vacuity, they are not the proof.

m	B	answer classes	fibres	acc	predicted
1	0	2	1	0.5000	0.5000
1	1	2	2	1.0000	1.0000
1	2	2	2	1.0000	1.0000
2	0	4	1	0.2500	0.2500
2	1	4	2	0.5000	0.5000
2	2	4	4	1.0000	1.0000
2	3	4	4	1.0000	1.0000
3	0	8	1	0.1250	0.1250
3	1	8	2	0.2500	0.2500
3	2	8	4	0.5000	0.5000
3	3	8	8	1.0000	1.0000
3	4	8	8	1.0000	1.0000
4	0	16	1	0.0625	0.0625
4	1	16	2	0.1250	0.1250
4	2	16	4	0.2500	0.2500
4	3	16	8	0.5000	0.5000
4	4	16	16	1.0000	1.0000
4	5	16	16	1.0000	1.0000

Table 26. All 18 (m, B) enumeration cells. Accuracy equals the predicted upper bound 2^B / |A| in every row and reaches 1 exactly at B = m.

m	B	layers	|A|	acc	output dev
2	0	1	4	0.2500	0.0e+00

4	4	2	16	0.0000	0.0e+00
4	4	3	16	0.0000	0.0e+00
4	4	4	16	1.0000	0.0e+00
4	4	5	16	0.0898	0.0e+00
2	2	3	4	1.0000	0.0e+00
2	2	4	4	1.0000	0.0e+00
3	3	4	8	1.0000	0.0e+00
3	3	5	8	1.0000	0.0e+00
4	4	5	16	1.0000	0.0e+00
4	4	6	16	1.0000	0.0e+00

Table 27. All 98 depth-scan cells. Accuracy and output deviation are invariant to layer count, which is the empirical form of Theorem A(e): depth cannot manufacture the missing bits.

carrier	bits / token	acc	Toeplitz	run-time tag
explicit tag	1	1.0000	yes	yes
codebook partition	1	1.0000	yes	no

Table 28. All 3 bit carriers. The position-format-prior accuracy 0.2969 is a SINGLE-SEED measurement on the alternating layout under that carrier's support assumption; it is a DIFFERENT configuration from the frozen five-seed parity-gapped value in Table 10 and is never quoted as a conclusion elsewhere. The position format prior is a support restriction, not a per-token channel, and is analysed through A(v) rather than through B; its effective bit count under arbitrary filler is zero.

hops	layers	acc	per-layer accuracies
1	1	1.0000	1.00
1	2	0.0000	1.00 0.00
1	3	0.0000	1.00 0.00 0.00
1	4	0.0000	1.00 0.00 0.00 0.00
1	5	0.0645	1.00 0.00 0.00 0.00 0.06
2	1	0.0000	0.00
2	2	1.0000	0.00 1.00
2	3	0.0000	0.00 1.00 0.00
2	4	0.0000	0.00 1.00 0.00 0.00
2	5	0.0684	0.00 1.00 0.00 0.00 0.07
3	1	0.0000	0.00
3	2	0.0000	0.00 0.00
3	3	1.0000	0.00 0.00 1.00
3	4	0.0000	0.00 0.00 1.00 0.00
3	5	0.0879	0.00 0.00 1.00 0.00 0.09
4	1	0.0000	0.00
4	2	0.0000	0.00 0.00
4	3	0.0000	0.00 0.00 0.00
4	4	1.0000	0.00 0.00 0.00 1.00
4	5	0.0918	0.00 0.00 0.00 1.00 0.09

Table 29. All 20 zero-training layer-by-hop cells with the full per-layer accuracy vector. The last-layer read-out is why H_L > H overshoots fail: the correct intermediate value is computed at layer H but is not the layer that is read out.

Appendix I Complete Numeric Provenance and Witness Detail

Every numeric value quoted anywhere in the paper is listed below with its statistic, seed count, sample count and the object it was measured on. This table exists so that no number has to be taken on trust and no earlier drifted value can re-enter unnoticed. All entries are finite-sample quantities unless the statistic column says otherwise; none is a theorem.

quantity	value	statistic	seeds / n	object
chance (this task)	0.0312	definition	-	1/V_K
H=2, 2 layers, 1500 steps	1.0000	mean / sd 0.0000	5 / 512	architecture stack
H=2, 1 layer, 1500 steps	0.1809	mean / sd 0.0219	5 / 512	architecture stack
H=2 paired difference	+0.8191	paired bootstrap	5	two-layer minus one-layer
paired 95%% CI	[+0.8008, +0.8375]	10000 resamples	5	same pair
first step >= 0.99	150	max over seeds	5	codebook-fixed interface
legacy random interface, 1500	0.2492	mean	5	learnable inter-layer matrix
SE2 predicted E[T_cover]	129.87	analytic n H_n / (1-beta)	-	n=32, beta~0
SE2 measured mean cover	130.00	mean over seeds	5	fused gated arm
SE2 relative gap	0.0987%	derived (mean coincidence, NOT precision)	-	prediction vs measurement
SE2 cover 95% t-CI	[82.72, 177.28]	t(4), sample SD 38.08	5	per-seed first full cover
SE3 exact E[acc]	0.709375	analytic	-	p=0.30, V_K=32
SE3 enumeration mean	0.709102	mean over 640000 key trials	20000 runs	exhaustive per-key
SE3 Hoeffding 4-sigma	[0.7071, 0.7116]	concentration	-	same
SE4 bits consumed	500	analytic	-	T=100, w=1, b=5
SE4 CT7 bound	500	analytic upper bound	-	same
SE5 K* at rho=0.999	6904	floor(log eps / log rho)	-	eps=1e-3
core state bytes	32768	exact	-	d x d fp64, all runs
fused final accuracy	1.0000	mean / sd 0.0000	5 / 512	320 rounds
ungated corrupted	0.6438	mean / sd 0.0781	5 / 512	p=0.30, no gate
break alignment	0.0250	mean / sd 0.0364	5 / 512	non-codebook registration
fast decay	0.1313	mean / sd 0.0234	5 / 512	rho=0.90
SE1 max on-target error	4.4e-16	max over 200 instances	200	orthogonality identity
SE1 max off-target error	8.5e-17	max over 200 instances	200	same
FFT/IIR duality (final state)	2.89e-16	max over probes	-	real-field equivalence
Toeplitz deviation (final)	0	exact	-	static recurrence

Table 30. All 26 provenance rows. Values from different probes are never averaged together; the FFT/IIR row is a maximum over separately listed probes, not a mixture.

witness	measured value	what it certifies
evolution all invariants	pass	consistency on final weights
evolution monotone	pass	consistency on final weights
fused above 0.90	pass	consistency on final weights
all key paired positive	pass	consistency on final weights
core state bytes constant across rounds	pass	consistency on final weights
slot count bounded by capacity	pass	consistency on final weights
SE5 capacity predicts rho fast loss	pass	consistency on final weights
ablation clean gate configs saturate	pass	consistency on final weights
ablation noisy gate necessary	pass	consistency on final weights
ablation rho decay matches SE5	pass	consistency on final weights
SE2 prediction matches measurement	pass	consistency on final weights
SE3 concentration holds	pass	consistency on final weights
SE3 enumeration within 4sigma	pass	consistency on final weights
no gradient through retrieval	pass	consistency on final weights
no NxN score	pass	consistency on final weights
no kv cache	pass	consistency on final weights
SE2 prediction gap	0.0987%	coupon-collector bound is non-vacuous
SE3 concentration	inside 4-sigma	exact expectation, not an upper bound
SE4 equality	500 = 500 bits	no contradiction with CT7
SE5 capacity	6904 vs 65	decay predicts the loss

Table 31. Witness detail for the non-analytic checks. A false here would be a finding to fix, not a crashed run.

hops	decoding	n	mean |A|	max log2|A|	frac |A|>1
1	flat	1280	1.00	0.00	0.00
1	codebook	1280	1.00	0.00	0.00
2	flat	1280	1.52	1.00	0.52
2	codebook	1280	1.00	0.00	0.00
3	flat	1280	2.36	2.00	0.75
3	codebook	1280	1.00	0.00	0.00
4	flat	1280	3.25	2.32	0.97
4	codebook	1280	1.00	0.00	0.00

Table 32. Compatible-answer-set enumeration for the original content-chain task. The bound B >= log2 |A(v)| is sufficient but not tight on this task; tightness belongs to the binary-chain witness of Appendix H. This distinction is stated in Section 4.3 and is not hidden here.

Appendix J Per-Seed Raw Curves at Key Steps

Appendix F averages over seeds at each step; this appendix prints the individual trajectories so that no seed-level failure can be hidden by a mean. Each row is one seed of one arm at one evaluation step. The five evaluation steps are 0, 150, 400, 800 and 1500, taken from the same 1500-step runs used in Section 4.12. Loss is the held-out cross-entropy on a fixed 64-instance probe batch; accuracy is on the full 512-instance held-out set.

arm	seed	step	acc	loss
arch codebook M	0	0	0.0273	3.4683
arch codebook M	0	150	1.0000	0.0655
arch codebook M	0	1500	1.0000	0.0017
arch codebook M	1	0	0.0273	3.4534
arch codebook M	1	150	1.0000	0.0608

arch codebook M	1	1500	1.0000	0.0015
arch codebook M	2	0	0.0234	3.4731
arch codebook M	2	150	1.0000	0.0597
arch codebook M	2	1500	1.0000	0.0014
arch codebook M	3	0	0.0371	3.4595
arch codebook M	3	150	1.0000	0.0751
arch codebook M	3	1500	1.0000	0.0024
arch codebook M	4	0	0.0176	3.4635
arch codebook M	4	150	1.0000	0.0756
arch codebook M	4	1500	1.0000	0.0021
arch analytic witness	0	0	1.0000	0.0000
arch analytic witness	1	0	1.0000	0.0000
arch analytic witness	2	0	1.0000	0.0000
arch analytic witness	3	0	1.0000	0.0000
arch analytic witness	4	0	1.0000	0.0000
legacy learnable interface rando	0	0	0.0332	3.4666
legacy learnable interface rando	0	150	0.2852	3.8409
legacy learnable interface rando	0	1500	0.2617	4.9533
legacy learnable interface rando	1	0	0.0293	3.4664
legacy learnable interface rando	1	150	0.2637	4.0069
legacy learnable interface rando	1	1500	0.2129	5.6684
legacy learnable interface rando	2	0	0.0371	3.4607
legacy learnable interface rando	2	150	0.2207	3.6956
legacy learnable interface rando	2	1500	0.2090	4.7915
legacy learnable interface rando	3	0	0.0215	3.4631
legacy learnable interface rando	3	150	0.3223	3.6256
legacy learnable interface rando	3	1500	0.2734	6.1103
legacy learnable interface rando	4	0	0.0293	3.4697
legacy learnable interface rando	4	150	0.3145	3.8663
legacy learnable interface rando	4	1500	0.2891	4.8512
legacy learnable interface fixed	0	0	0.0391	3.4783
legacy learnable interface fixed	0	150	1.0000	0.1207
legacy learnable interface fixed	0	1500	1.0000	0.0019
legacy learnable interface fixed	1	0	0.0215	3.4747
legacy learnable interface fixed	1	150	1.0000	0.0738
legacy learnable interface fixed	1	1500	1.0000	0.0017
legacy learnable interface fixed	2	0	0.0312	3.4700
legacy learnable interface fixed	2	150	1.0000	0.0581
legacy learnable interface fixed	2	1500	1.0000	0.0015
legacy learnable interface fixed	3	0	0.0449	3.4793
legacy learnable interface fixed	3	150	1.0000	0.1008
legacy learnable interface fixed	3	1500	1.0000	0.0027
legacy learnable interface fixed	4	0	0.0566	3.4705
legacy learnable interface fixed	4	150	1.0000	0.0731
legacy learnable interface fixed	4	1500	1.0000	0.0021
legacy trainable near-swap	0	0	0.0234	3.4641
legacy trainable near-swap	0	150	1.0000	0.0569
legacy trainable near-swap	0	1500	1.0000	0.0016
legacy trainable near-swap	1	0	0.0410	3.4781
legacy trainable near-swap	1	150	1.0000	0.0623
legacy trainable near-swap	1	1500	1.0000	0.0016
legacy trainable near-swap	2	0	0.0254	3.4697
legacy trainable near-swap	2	150	1.0000	0.0496
legacy trainable near-swap	2	1500	1.0000	0.0013
legacy trainable near-swap	3	0	0.0215	3.4606
legacy trainable near-swap	3	150	1.0000	0.1082
legacy trainable near-swap	3	1500	1.0000	0.0025
legacy trainable near-swap	4	0	0.0078	3.4713
legacy trainable near-swap	4	150	1.0000	0.0681
legacy trainable near-swap	4	1500	1.0000	0.0022

Table 33. All 65 per-seed curve values. The legacy random-interface rows stay between 0.21 and 0.29 at every step on every seed, far from this task's 1/32 chance and never above 0.30; the codebook-fixed rows reach 1.0 by step 150 on every seed and never regress. The analytic witness is exact at step 0. No seed was excluded and no value was rounded from a different run.

arm	seed	first full cover	committed	core bytes
fused gated	0	170	49	32768
fused gated	1	100	55	32768
fused gated	2	90	47	32768
fused gated	3	120	53	32768
fused gated	4	170	54	32768
no consolidation	0	none	0	32768
no consolidation	1	none	0	32768
no consolidation	2	none	0	32768
no consolidation	3	none	0	32768
no consolidation	4	none	0	32768
ungated clean	0	170	320	32768
ungated clean	1	100	320	32768
ungated clean	2	90	320	32768
ungated clean	3	120	320	32768
ungated clean	4	170	320	32768
ungated corrupt	0	none	320	32768
ungated corrupt	1	none	320	32768
ungated corrupt	2	none	320	32768
ungated corrupt	3	none	320	32768
ungated corrupt	4	none	320	32768
random write	0	none	154	32768
random write	1	none	153	32768
random write	2	none	154	32768
random write	3	none	155	32768
random write	4	none	155	32768
break alignment	0	none	49	32768
break alignment	1	none	55	32768
break alignment	2	none	47	32768
break alignment	3	none	53	32768
break alignment	4	none	54	32768
rho fast	0	none	49	32768
rho fast	1	none	55	32768
rho fast	2	none	47	32768
rho fast	3	none	53	32768
rho fast	4	none	54	32768

Table 34. All 35 fusion runs with their first full-cover round, number of committed experiences and core state size. The core state is constant across every arm, every seed and every round; only the bounded slot metadata changes. These are the individual values behind Table 16 and Table 18.

End of artifact. No further numeric claims appear anywhere in the submission.